

Inert Gas Transport in Blood and Tissues

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ABSTRACT

This article establishes the basic mathematical models and the principles and assumptions used for inert gas transfer within body tissues—first, for a single compartment model and then for a multicompartment model. From these, and other more complex mathematical models, the transport of inert gases between lungs, blood, and other tissues is derived and compared to known experimental studies in both animals and humans. Some aspects of airway and lung transfer are particularly important to the uptake and elimination of inert gases, and these aspects of gas transport in tissues are briefly described. The most frequently used inert gases are those that are administered in anesthesia, and the specific issues relating to the uptake, transport, and elimination of these gases and vapors are dealt with in some detail showing how their transfer depends on various physical and chemical attributes, particularly their solubilities in blood and different tissues. Absorption characteristics of inert gases from within gas cavities or tissue bubbles are described, and the effects other inhaled gas mixtures have on the composition of these gas cavities are discussed. Very brief consideration is given to the effects of hyper- and hypobaric conditions on inert gas transport. © 2011 American Physiological Society. *Compr Physiol* 1:569–592, 2011.

Introduction

In the context of this chapter, inert gases refer to metabolically inactive gases that are either inhaled naturally such as nitrogen and rare noble gases in the air, inhaled for measurement purposes such as sulphur hexafluoride (SF₆), acetone and radioactive gases, or inhaled during anaesthesia such as nitrous oxide (N₂O) and volatile halogenated agents. Although oxygen has been shown to be used in certain conditions as an insoluble “inert” gas similar to argon in respiratory sinusoidal forcing measurements (35, 84), this will not be discussed further. The various areas where inert gas transport is currently of interest are listed in Table 1, though not all areas will be considered in detail.

As pointed out in previous editions of the *Handbook of Physiology*, which were forerunners to this text, there has been a long history of modeling of inert gas exchange and the general theoretical basis for the models developed has been extensively discussed there together with a list of previous studies from the early work of Zuntz (88) together with Kety (45), Mapleson (52), Eger (19), and Zwart et al. (89), to list some of the more famous names (76).

Basic Principles

Models and algebraic analysis

A single-compartment model with constant arterial partial pressure

In its simplest form, a single tissue element perfused by a single capillary, whose arterial inert gas partial pressure is constant, may be considered. Such a model may be expanded

to encompass more physiologically realistic situations where the body comprises an array of different tissues perfused in parallel and where the arterial inert gas partial pressure is not constant but is influenced by venous recirculation and the ventilation-perfusion ratio of the lung. For all models, however, the following assumptions are required:

- The tissue element is spatially homogeneous, both axially along the length of its interface with the capillary and radially, that is, perpendicular to the axial plane at all tissue depths.
- The inert gas partial pressure in the capillary, P_c , is homogeneous along the capillary, is instantaneously in equilibrium with the tissue partial pressure, P_{ti} , throughout its course and equal to the venous effluent partial pressure, P_v .
- In a multitissue compartment model, gas exchange occurs only between an individual compartment and its capillaries. No diffusion occurs between compartments; that is, no “leakage shunting” occurs.
- Capillary volume is negligibly small compared to tissue volume.

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Published online, April 2011 (comprehensivephysiology.com)

DOI: 10.1002/cphy.c100011

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Table 1 Inert Gas Transport

Study of fundamental factors of inert gas transport in the lung and peripheral tissues
Basic principles and assumptions of models of uptake and elimination of inert gases
Multicompartment considerations (functional heterogeneity within/between tissues)
Equilibrium and absorption in body cavities and gas bubbles
Roles of diffusion and of perfusion in gas exchange
Perfusional and diffusional tissue “shunts”
Hyperbaric and hypobaric considerations
Decompression consequences
Narcosis
Central nervous system effects
Effect on body cavities
Anesthetic considerations
Quantative factors in anesthetic uptake and elimination
Applications of tissue inert gas exchange
Estimation of tissue compartment volumes
Estimation of organ perfusion
Radioactive gas tissue dosimetry

Accepting these assumptions, simple mass balance principles may be applied to the delivery, uptake, and elimination of an inert gas with respect to a single tissue element, as shown in Figure 1, whose arterial tension is constant but whose tissue and venous tensions rise and are functions of time.

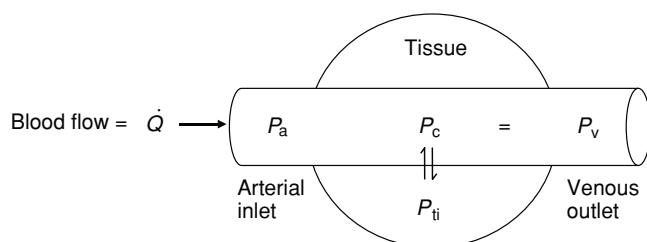
The tissue volume is V_{ti} . The solubility of gas in tissue is S_{ti} , and the solubility of gas in blood is S_b . The mass of gas dissolved in the tissue is therefore equal to $V_{ti}P_vS_{ti}$. The ratio S_{ti}/S_b is the tissue-gas partition coefficient, λ .

The rate of delivery of tracer gas to the tissue via the arterial blood is $\dot{Q} P_a S_b$. The rate of gas loss in the venous effluent is $\dot{Q} P_v(t) S_b$. The difference between these rates must equal the rate of change of mass of gas in the tissue compartment. Hence,

$$\dot{Q} S_b (P_a - P_v(t)) = \frac{d(V_{ti} P_{ti}(t) S_{ti})}{dt} \quad (1)$$

Equation (1) can be simplified by letting the tissue and venous partial pressure terms be equal and by acknowledging that the tissue volume and solubility terms are constant. Hence,

$$\dot{Q} S_b (P_a - P_v(t)) = V_{ti} S_{ti} \frac{dP_v(t)}{dt} \quad (2)$$

**Figure 1** Single element capillary-tissue model.

Equation (2) is a first-order differential equation that can be rearranged, to allow integration, as shown below:

$$\int_0^{P_v} \frac{dP_v(t)}{(P_a - P_v(t))} = \frac{\dot{Q} S_b}{V_{ti} S_{ti}} \int_0^t dt$$

The solution to this standard integral gives

$$[\ln(P_a - P_v(t))]_0^{P_v} = - \left(\frac{\dot{Q} S_b}{V_{ti} S_{ti}} \right) [t]_0^t$$

applying the limits such that when $t = 0$, $P_v = 0$, gives:

$$\ln(P_a - P_v(t)) - \ln(P_a) = - \left(\frac{\dot{Q} S_b}{V_{ti} S_{ti}} \right) t$$

$$\ln \left(\frac{P_a - P_v(t)}{P_a} \right) = - \left(\frac{\dot{Q} S_b}{V_{ti} S_{ti}} \right) t$$

raising both sides to the power “e” gives:

$$\frac{P_a - P_v(t)}{P_a} = \exp \left[- \left(\frac{\dot{Q} S_b}{V_{ti} S_{ti}} \right) t \right]$$

Hence,

$$P_v(t) = P_a \left[1 - \exp \left[- \left(\frac{\dot{Q} S_b}{V_{ti} S_{ti}} \right) t \right] \right] \quad (3)$$

Equation (3) describes an exponential wash-in function in which the tissue partial pressure starts at zero and rises to its maximum value, P_a , at $t = \infty$. The rate constant of this exponential wash-in process is given by the term in the rounded bracket, that is, $\dot{Q} S_b / V_{ti} S_{ti}$. It can be seen that the numerator of this term is the product of blood flow and blood solubility and represents the “conductance” of the system, that is, how much flux can be delivered by the blood per unit of partial pressure. The denominator of the rate constant term is the product of the tissue volume and tissue solubility and represents the “capacitance” of the system, that is, how much mass can be absorbed by the tissues per unit of partial pressure. It is therefore the conductance:capacitance ratio of a tissue element that determines its wash-in rate constant. This confirms the intuitive notion that tissues with a large blood flow relative to their volume will have rapid wash-in kinetics.

Also of note is that the rate constant term can be simplified by substituting the ratio of the tissue-gas and blood-gas solubilities, S_{ti}/S_b , with the tissue-blood partition coefficient, λ , to give the following expression:

$$P_v(t) = P_a \left[1 - \exp \left[- \left(\frac{\dot{Q}}{V_{ti} \lambda} \right) t \right] \right] \quad (4)$$

Equation (4) shows us that the conductance:capacitance ratio (i.e., the exponential rate constant) is dependent on the tissue-blood partition coefficient and is seemingly independent of the individual tissue and blood solubility terms. This

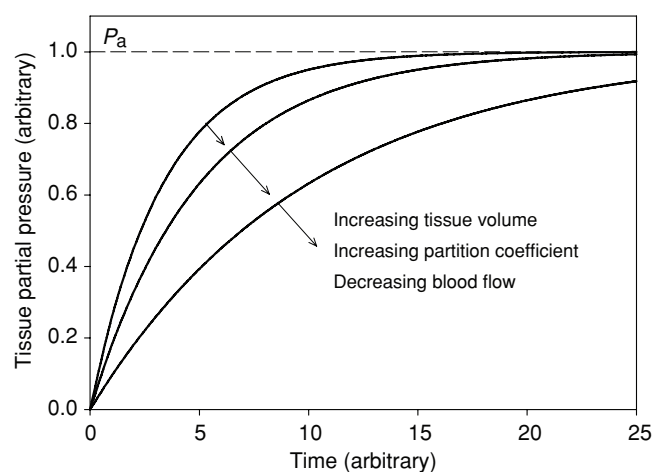


Figure 2 Exponential wash-in of inert gas in a perfused single tissue element. The tissue (and venous) partial pressure rises from zero asymptotically to arterial partial pressure. The rate constant is reduced by increased tissue volume, increased tissue-gas partition coefficient, and decreased blood flow.

produces a somewhat unintuitive outcome, namely, that tissues with high capacitance, due to high tissue solubility of a particular inert gas, may not necessarily demonstrate slow wash-in kinetics if at the same time the inert gas has similarly a high blood solubility (i.e., a high conductance). This becomes clearer when one considers that the “slowing effect” of the high tissue capacitance (due to high tissue-gas solubility) is offset by the “quickening effect” of the high conductance or delivery of the inert gas to the tissue (due to the high blood-gas solubility). The wash-in kinetics for this single-compartment model are shown in Figure 2.

Time dependency models

This ordinary differential equation (ODE) model does not fully consider the time-dependent element which is relevant to dynamic situations with fluctuating input, as in the induction of anesthesia or sine-wave measurement techniques. In these situations, a partial differential equation (PDE) model is more complete, incorporating convection and diffusion elements, particularly with respect to gas transfer in the airways that affects the inert gas load delivered to the alveolar-capillary interface and subsequently the capillary-tissue interface (82). The ODE model responds in a damped manner when compared with the PDE model, because the PDE model more efficiently processes the simultaneous aspects of gas diffusion and convection in the airways, capillaries, and tissues. The results for venous $P_v(t)$ following a step change in $P_a(t)$ are shown in Figure 3 (82), demonstrating that although the effects of PDE over ODE models with respect to convection and diffusion are potentially more marked within the airways, these effects are also of importance in the tissues. Thus, the studies using the ODE model (the majority of the published literature) will result in inaccuracies in dynamic situations when compared to steady-state situations, though there will

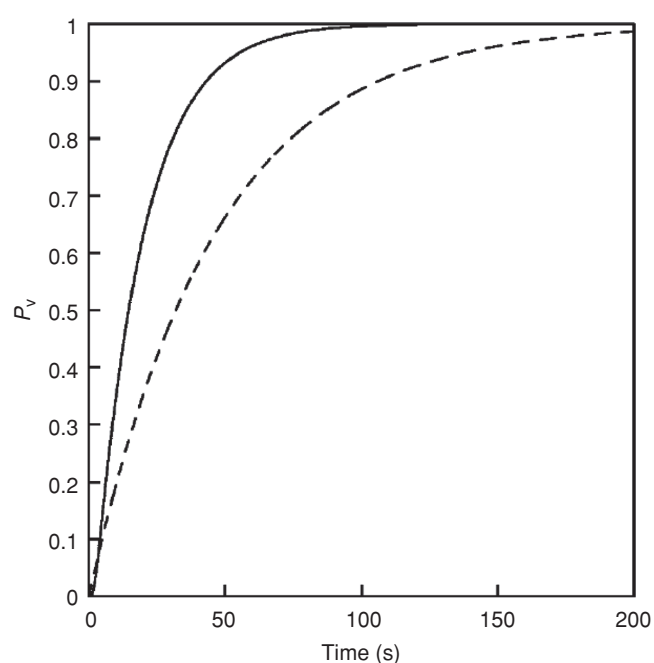


Figure 3 $P_v(t)$ calculated using both the PDE model (solid line) and the ODE model (broken line) for the uptake of the anesthetic gas nitrous oxide where $P_a(t)$ has a step change from 0 to 1 at time $t = 0$. Showing the more rapid response in the PDE model. From Whiteley et al. (82), with permission.

be no difference in total gas exchange or in the general relativities of responses with differing inert gases at steady state. Equilibration will, however, be quicker than previously predicted (19), though the results also depend on the assumed size of the body compartment model (82).

A multicompartment model with constant arterial partial pressure

This simple single-compartment model may now be expanded to include a number of different tissues whose blood flows and tissue-blood partition coefficients can be defined. Amongst the first multicompartmental analyses was that of Kety (44,45), who described how inert gases were driven into a series of parallel tissue compartments by the gas tension gradient existing between the arterial blood and the compartment under consideration. In man, three or four tissue groups have been defined for the purposes of modeling the uptake and elimination of volatile anesthetic agents (31, 53, 89). These body compartments have come to be known as:

- the “vessel-rich group” (VRG), comprising the heart, kidney, liver, and brain
- the muscle group (MG)
- the fat group (FG)
- the “vessel-poor group” (VPG), comprising avascular structures such as tendon and other connective tissues.

Table 2 Blood Gas Solubilities (S_b) and Blood Tissue Partition Coefficients (λ) for Different Inert Gases

		VRG	MG	FG	VPG
$\dot{Q}$, liter/min		4.5	1.1	0.32	0.08
V_{ti} , liter		6	36	15.2	13
Gas or vapor	S_b , ml/ml/atm	λ , unitless			
SF ₆	0.008 ^a				
Helium	0.009 ^b				
Nitrogen	0.0146	1.1	1.2	5.2	1.0
Argon	0.029 ^b				
Ethane	0.088 ^a				
Xenon	0.14 (0.115 ^c)	1.43	0.71	12.9	
Desflurane	0.42 ^d	1.29 ^d	2.31 ^d	30.9 ^d	
Nitrous oxide	0.47	0.8	1.17	2.3	1.0
Cyclopropane	0.5 (0.36 ^a)	1.2	1.32	28	1.2
Sevoflurane	0.6 (0.65 ^d)	2.17 (1.69 ^d)	2.5 (3.69 ^d)	71.7 (52.3 ^d)	
Isoflurane	1.4 (1.3 ^d)	2.07 (1.62 ^d)	1.71 (3.46 ^d)	49.3 (53.8 ^d)	
Enflurane	1.8	1.83	2.67	73.3	2.0
Halothane	2.4 (2.5 ^d)	2.46 (1.92 ^d)	3.46 (3.96 ^d)	92.5 (60 ^d)	3.0
Chloroform	8	2.5	1.5	35	
Diethyl ether	12 (15.2 ^a)	1.08	0.83	5.83	1.0
Methoxyflurane	14	1.93	1.93	63.2	
Toluene	10 (15.6 ^a)	3.6	3.5	96.2	
Acetone	330 ^e (341 ^a)	0.45 ^e	0.45 ^e	0.26 ^e	
Ethyl alcohol	~1750 ^f (2516 ^e)	0.41 ^e	0.34 ^e	0.09 ^e	

Data from Fiserova-Bergerova and Diaz (26).

^aAnderson and Hlastala (4).

^bBurkard and Van Liew (13).

^cGoto et al. (32).

^dEger and Shafer (21).

^ePaterson and Mackay (58).

^fHlastala (37).

VRG, vessel-rich group; MG, muscle group; FG, fat group; VPG, vessel-poor group; $\dot{Q}$, organ blood flow; V_{ti} , organ volume.

The volumes and blood flows of these compartments, and their partition coefficients for a range of inert gases, are shown in Table 2.

These tissue groups are connected in parallel and supplied by a common arterial input. The venous effluents of these compartments join to form a common mixed venous flow, and the inert gas content and partial pressure of this common mixed venous blood are determined by the flow-weighted partial pressures of the venous effluent from each individual compartment. So for n compartments, mixed venous partial pressure, $P\bar{v}$, is given by:

$$P\bar{v} = \frac{\sum_{j=1}^n (P_{v,j} \dot{Q}_j)}{\sum_{j=1}^n \dot{Q}_j} \quad (5)$$

We have neglected to consider the transit times for the venous recirculation of each compartment, j , assuming for simplicity that each contributes synchronously and instantaneously to the mixed venous partial pressure. In many cases, this is not an unreasonable assumption since the transit time is

considerably less than the exponential time constant. When circulation time is ignored, Mapleson (52) has shown that significant errors are confined approximately to the first 2 min of uptake or elimination. The schematic arrangement is shown in Figure 4.

Equation (4) and the data from Table 2 allow us to determine the tissue partial pressure of an inert gas in each of the body compartments at any time after the imposition of a constant arterial partial pressure, and Eq. (5) allows us also to determine the mixed venous partial pressure. These are plotted in Figure 5 for the anesthetic vapors desflurane and diethyl ether.

For both vapors, the VRG is seen to equilibrate rapidly with the arterial blood, because the VRG, which receives 75% of the cardiac output, accounts for only 9% of the total body volume. The speeds of rise and fall of the mixed venous partial pressures are also quite similar and rapid as predicted from Eq. (5), since the mixed venous partial pressure is heavily weighted by the VRG.

Desflurane and ether occupy opposite ends of the spectrum of anesthetic blood-gas solubilities and have markedly different clinical characteristics. Desflurane is a modern agent with a low blood-gas solubility ($S_b = 0.42$), which allows rapid

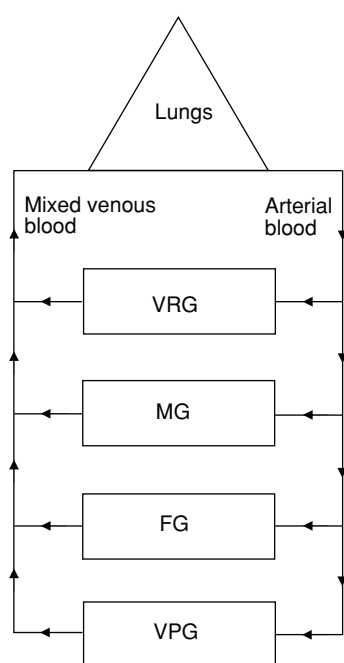


Figure 4 Schematic representation of body compartments. VRG, vessel-rich group; MG, muscle group; FG, fat group; and VPG, vessel-poor group.

induction of anesthesia. Ether, in contrast, is an older agent with higher blood-gas solubility ($S_b = 12.0$) and is associated with markedly slower induction. Figure 5A, however, shows that despite the markedly different clinical and physicochemical properties of these two vapors, the rates of rise and fall of partial pressure in the VRG (i.e., the brain) are very similar

in the above model because their tissue-blood partition coefficients, are so similar (see Table 2), and it was shown in Eq. (4) that it is λ that determines the wash-in and washout rate kinetics. How then can we explain the observed physiological and clinical differences between these agents? We address this question in the Anesthetic Uptake and Elimination section (context sensitivity).

Observation of Figure 5 also reveals an apparent asymmetry between the wash-in and washout phases, which becomes more pronounced in the increasingly slower compartments; compare the VRG, which is almost symmetrical, with the FG, which is very asymmetrical. This is because the “driving pressures” during wash-in and washout are very different. For example, consider the FG in Figure 5. During wash-in, the “driving pressure,” or “pressure head,” responsible for driving gas into the FG is approximately 0.95 pressure units at 60 min. At 61 min, in the early washout phase, the “driving pressure” responsible for driving gas out of the FG is the difference between tissue pressure and zero, that is, approximately 0.05 units. In contrast, in the VRG, the driving pressures are the same for both wash-in and washout.

Intertissue diffusion

In the model so far, all vapor within a tissue compartment has been delivered there directly by the blood perfusing that compartment. However, this is likely to be a simplification of reality.

In 1965, Rackow and colleagues (63) studied the simultaneous uptake of N_2O and cyclopropane in humans. These inert gases have almost identical blood solubility and solubilities in “watery” compartments such as muscle and viscera,

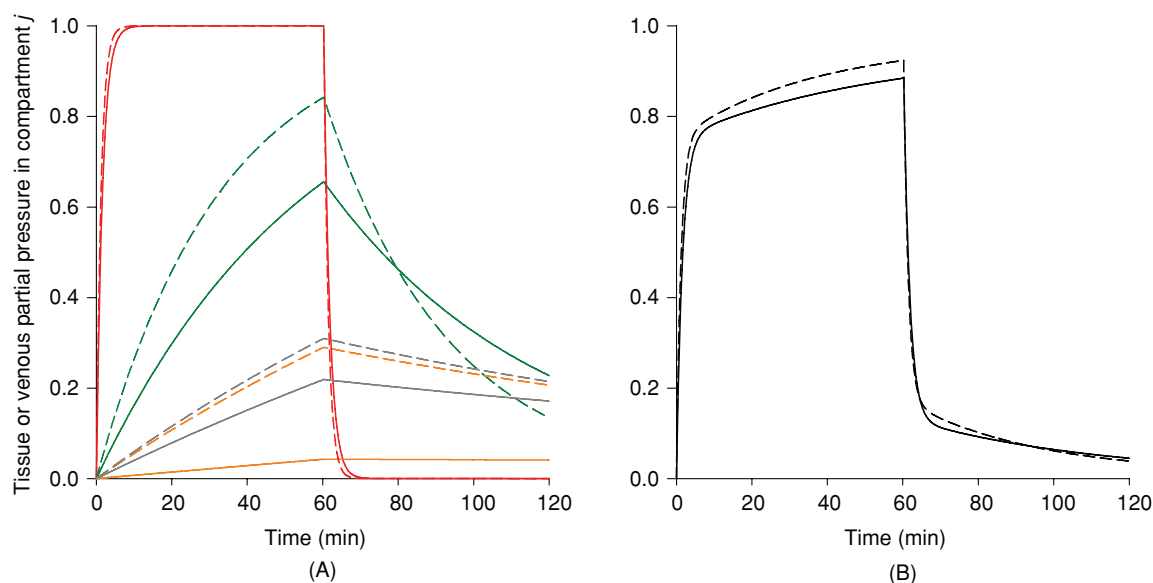


Figure 5 (A) Compartmental venous and tissue partial pressure versus time after imposition of a constant arterial partial pressure of 1.0. At 60 min, the arterial partial pressure is set to zero. Red lines, vessel-rich group; green lines, muscle group; gray lines, vessel-poor group; and orange lines, fat group. Solid lines represent the anesthetic vapor desflurane, dotted lines represent ether. (B) The mixed venous partial pressures for the same simulation.

yet their solubility in fat differs considerably. Given the low perfusion of the fat compartment, one might expect to see very little difference in the uptake rates of these two gases, yet even in the early stages of the wash-in, the alveolar partial pressure of cyclopropane was consistently around 5% less than that of N₂O. Clearly, the capacitance of the fat compartment for cyclopropane was exerting an effect that was disproportionate to its conductance, and this is inconsistent with the simple model described above. Perl and colleagues (59) provided an explanation as to how the capacitance of the fat compartment could be charged disproportionately to its conductance by suggesting that some gas delivered to the VRG (which enjoys high conductance) transfers to adjacent fat by intertissue diffusion. This means that a part of the VRG is slower to reach equilibrium because gas is continuously lost to adjacent fat. This phenomenon gives rise to the concept of another tissue group, namely, the “intertissue diffusion group,” with its own volume, blood flow, and time constants (20).

Because of the multicompartmental exponential nature of the uptake or elimination curves for gases, it is not possible to determine whether diffusion rates have a measurable effect on tissue-gas exchange. Also, it is logical that tissues are unlikely to be spatially uniform and this fact, together with the random orientation of capillaries, arterioles and venules, means that different regions will have different diffusion rates, which will further confuse the interpretation of measured data. With respect to diffusion limitation and wash-in or washout time, Kety (45) concluded that equilibrium between blood and tissue is essentially complete within 1 s, though Forster (28) calculated a $t_{1/2}$ of ~55 s for resting skeletal muscle.

Square root of time relationship

Severinghaus (69) studied the uptake of N₂O in volunteers, and being versed in the exponential nature of the process, expected to see a straight line relationship between uptake and time when the data were plotted on semilogarithmic paper. However, this was not the case, and only when plotted on double logarithmic axes was a straight line relationship revealed. Severinghaus noted that the gradient of this plot was -0.5 , implying that the uptake was inversely proportional to $\sqrt{t}$. This empirical black-box description has no basis in physiological modeling, yet is compellingly robust and gives seemingly identical predictions to multicompartment exponential models as described above. Connor and Philip (16) examined the relationship between the behaviors of the multicompartment and square root of time models in a theoretical study designed to assess the following equivalence, where the multicompartment model uptake is shown on the left-hand side and the Severinghaus uptake on the right:

$$\sum_{j=1}^n a_j e^{-\frac{t}{\tau_j}} = A t^k$$

The value of A was set at 1 liter/min and the value of k at -0.5 (although errors in k of $\pm 20\%$ were also tested).

Using a numerical iterative method to determine which values of a_j and τ_j for n compartments best support the equivalence, it was shown that while a single-compartment model failed to match the Severinghaus curve, both three- and four-compartment models fitted the Severinghaus curve almost indistinguishably. The time constants of these compartments were relatively insensitive to variations in k value, but what is particularly interesting is that the best-fitted τ_j values for the three-compartment model are very close to those corresponding to real physiological compartments. For example, the fastest compartment had a time constant of 1.46 min, which corresponds almost exactly to that of the VRG (1.4 min), and the slowest had a time constant of 42 min, which is strikingly similar to that quoted for the muscle group (42.1 min). Connor and Philip considered the identity of the middle compartment ($\tau = 7.42$ min) to be somewhat more mysterious, since it did not correspond to any known physical compartment. Possible explanations are that this represents the time constant for intertissue diffusion within the brain or that since the VRG represents a number of different tissues (brain, kidney, heart), their slightly different time constants have been “dissected out” by the model. A final possibility is that their iterative technique has produced solutions that bear just coincidental resemblances to real compartmental time constants.

Steady-state versus dynamic models

Steady state versus dynamic

Inert gas uptake or elimination has for many years formed the basis of a measurement technique investigating the ventilation-perfusion inequality potentially existing at the lung-circulation interface. The theory, on which this measurement is based, depends on achieving a steady-state equilibrium or quasi-equilibrium state after 30 to 45 min of breathing or infusing constant concentrations of both soluble and insoluble inert gases, with a wide range of blood-gas solubilities from SF₆ to acetone, modeled in a 50-compartment lung (24, 77-79). This topic is more fully discussed elsewhere in *Comprehensive Physiology*.

The steady state does not allow consideration of dynamic responses to gas uptake or elimination, particularly for soluble inert gases and particularly in situations where there is lung pathology such as emphysema or pulmonary embolism. Recently, Yem and colleagues have produced a dynamic model (85) that allows for ventilation-volume ($\dot{V}_A/V_A$) effects to be incorporated as well as ventilation-perfusion ($\dot{V}_A/\dot{Q}$) effects. The 50-compartment lung model has achieved iconic recognition with zero $\dot{V}_A/\dot{Q}$ or “shunt” and infinity $\dot{V}_A/\dot{Q}$ or anatomical dead space as compartments together with another 48 additional parallel alveolar compartments representing a range of $\dot{V}_A/\dot{Q}$ ratios between 0.005 and 100.0 divided equally on a logarithmic scale (79). Although this model may

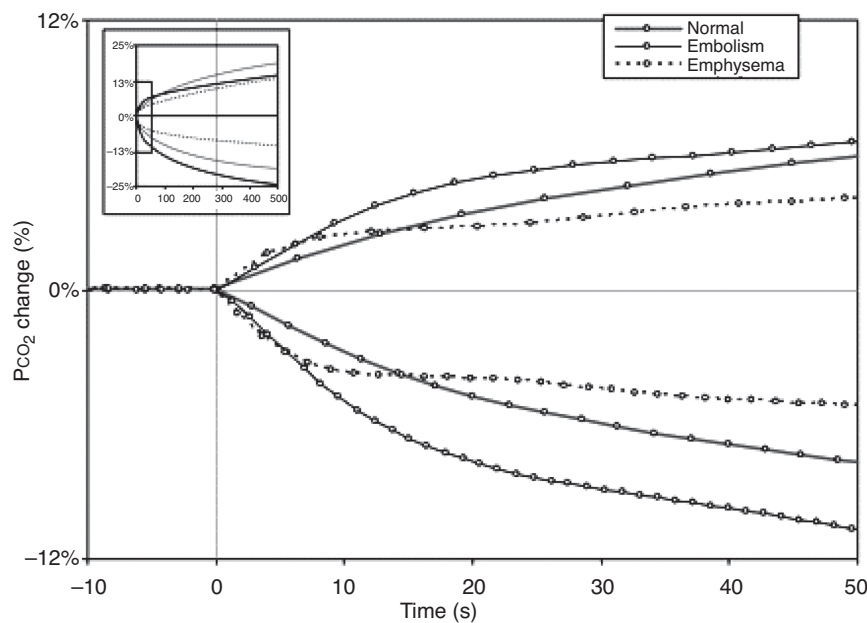


Figure 6 The simulated arterial PCO₂ responses to step changes in ventilatory frequency. The ventilatory frequency was increased or decreased by a factor of 1.5 for each model at $t = 0$. Gray line, normal lung model; black line, embolism model; and dashed line, emphysema model. The open circles represent end-tidal points. Main figure –10 to 50 s. Inset figure 0 to 500 s. From Yem et al. (85), with permission.

be used to produce a smoothed distribution of ventilation-perfusion inequality, such distributions have been shown to contain only a maximum of three distinct $\dot{V}_A/\dot{Q}$ modes, or two modes in addition to shunt, and dead space (79, 80). Thus, the ventilation-perfusion model may be reduced to three $\dot{V}_A/\dot{Q}$ compartments plus shunt and dead space compartments as published by a number of authors (47, 65, 66, 85). To incorporate a dynamic element into this model, gas washout information from the lung may be used, which, at best, allows for a maximum of two (or possibly three) $\dot{V}_A/\dot{V}_A$ compartments (23, 42, 49, 57). Combining these two simplifications allows a model with eight alveolar compartments (only a maximum of six of which may be operational for any individual case), and a shunt compartment (85).

Using this model, it has been shown that there are some differences in the response to a step change in ventilation for different pathophysiological states of the lungs. As shown in Figure 6, the initial response in both emphysema and embolism is faster than the normal and remains faster up to ~1 min for the embolism situation (particularly for a reduction in ventilation) (85). This model indicates a very early (<10 s) faster response for emphysema, with effects of recirculation, and the slower $\dot{V}_A/\dot{V}_A$ compartment later changing the response to a much slower continuing change. The response to a step change is much prolonged in emphysema, and measurement after a 30- to 45-min “stabilization” period may be while there is still change occurring as equilibration may not have occurred.

Other models have also indicated that dynamic situations may well be different from steady-state ones. Whiteley et al.

(83) have demonstrated that the type of ventilation pattern changes the way in which the lungs respond particularly with more detailed lungs models, though Nye (57) and Baker et al. (6) have demonstrated similar responses with much simpler models. To date, there have been no studies, in abnormal lungs, on the effects of dynamic ventilation on inert gases uptake and elimination, though it is to be expected that there would be important differences from the steady-state situation. This would have implications with anesthetic gases and volatile agents for subjects and patients with abnormal lungs, as steady state may take longer to establish.

It is possible that measurement of simple PaCO₂ change over a short period of time, such as 5 min, for a step change in ventilation may be able to provide a simple description of the degree of functional $\dot{V}_A/\dot{Q}$ mismatch when linked to a real-time computer model simulation of various potential lung conditions.

Airways models

Airways

Although this article concerns inert gas transport in tissues, it is pertinent to discuss lung uptake since this is usually the means whereby inert gases are introduced into the body. The “fish-gill” model for the lung predicts reasonably well experimental data, but more recently other lung models have been suggested and only rudimentary application of these models has so far been published. The “trumpet” model (83) has not yet been integrated with body compartments, and the

“balloon-on-a-straw” model (36) has been used in a few studies (81). Hahn and Farmery (36) highlighted certain studies where the “fish-gill” model was not a good predictor of the within-breath oscillations observed by Bergman (11), or later by Purves (62) and Folgering et al. (27), and more recently by Baumgardner et al. (9).

For soluble inert gases ($S_b > 100$), it is necessary to consider the exchange of gas in the proximal airways as well as the more usual alveolar-capillary exchange. When $S_b > 1000$, exchange is almost exclusively in the conducting airways because of gaseous dissolution in, and evolution from, bronchial mucous/water during tidal breathing in a way similar to that in which heat and water are exchanged between the airway and the environment (3, 37). This is the case for ethyl alcohol ($S_b \sim 1750$) (3, 37). Gases with $S_b < 10$ exchange almost entirely in the alveolar region since the airways simply act as passive conduits. The exchange of high-solubility gases in the airways is influenced by the bronchial blood flow, which is increased in conditions such as asthma, bronchiectasis, and smoke inhalation and may be increased in exercise. Exchange is also influenced by factors determining the rate of diffusion such as the molecular weight of the gas, the thickness of the tissue barrier (which may be affected by pathology such as bronchial infection or inflammation), and the temperature of the airway mucous/water. During exercise hyperpnea and in cold dry environmental conditions, airway gas may not become isothermal until considerably beyond the 10th airway generation, thereby increasing the capacity of the airways for soluble gas exchange. Soluble inert gases with $S_b < 1000$ exchange more readily in these proximal airways in such situations because the lower temperature increases the S_b of the inert gas when compared with the normal body temperature of 37°C, possibly by as much as 150% (3). The normal airways gas exchange is shown in Figure 7 for cyclopropane ($S_b \sim 0.5$), ether ($S_b \sim 12$), and acetone ($S_b \sim 330$). Anderson and colleagues have shown that for high-solubility gases, the end-exhaled concentration is significantly lower than the alveolar concentration and that the decrement depends on the rate of exhalation. Following a large inhalation (1.5 liters), the end-expiration concentration for acetone was 20% less than the alveolar concentration when the exhalation rate was slow (200 ml/s) and 15% less when fast (350 ml/s). (4). During tidal breathing, the decrement may be up to 30% less than the alveolar concentration. This research has implications both for the standardization of acetone measurements and for ethyl alcohol breath tests.

Density/viscosity and convection/diffusion effects

Diffusion may occasionally be against the axial convective flow. For example, nitric oxide (NO) may exhibit counter-convective diffusion because hemoglobin at the alveolar-capillary interface acts as a sink for NO and so NO in the terminal airways will diffuse down a concentration diffusion gradient toward the alveolus in the opposite direction to the exhalational axial flow (70). However, the diffusion gradient

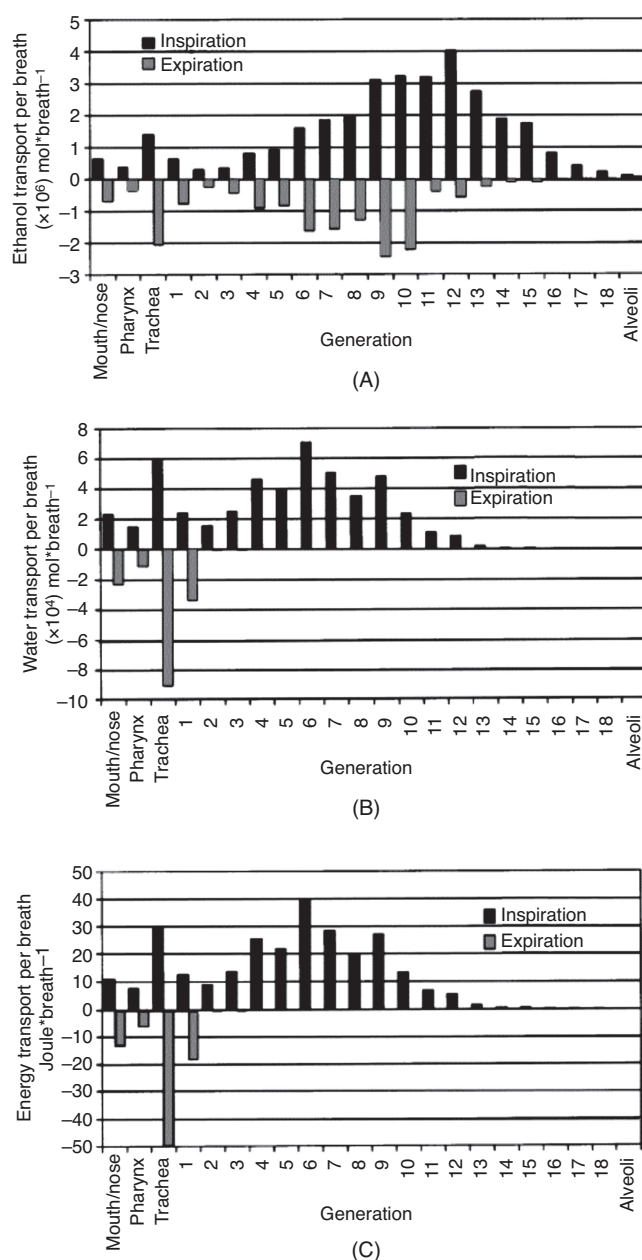


Figure 7 Axial distribution of gas transport during inspiration (black columns) and expiration (gray columns) for a tidal breath of cyclopropane (A), ether (B), and acetone (C). Each flux has been normalized by the total inspiratory soluble gas flux. As the blood solubility of gas increases from cyclopropane to acetone, the distribution shifts from a sharp concentrated peak in the alveolar region for cyclopropane to a wider distribution that spreads throughout the airways for acetone. From Anderson et al. (2), with permission.

will in most cases be in the same direction as the convective flow. In both situations, the diffusion flow will be small in relation to the axial convective flow and only of importance when the axial flow is low, or in breath-holding.

Dead space changes with different gases

The standard Bohr-Engelhof (12, 22) dead space measurement uses CO₂ as the marker gas to determine the dead space

volume. When other gases are considered, there will be a different dead space volume, which for some conditions will differ from that calculated using CO_2 . For example, in the MIGET technique for measuring $\dot{V}_A/\dot{Q}$ inequality, there is quite a wide solubility range for the inert gases used from ~ 0.01 (for SF_6) to ~ 330 (for acetone). For normal lungs with substantially homogenous ventilation, there is minimal ventilation to areas with $\dot{V}_A/\dot{Q}$ substantially greater than unity so that dead spaces calculated using CO_2 ($S_b \sim 4$) or acetone ($S_b \sim 330$) will not be appreciably different. However, for pathological lungs (e.g., with emphysema), with increased $\dot{V}_A/\dot{Q}$ substantially greater than unity but unlikely greater than 100, there will be a difference for the Bohr-Engdoff measured dead space, which will be greater than for the acetone dead space (25, 34, 85). The composition of gas in the common dead space of the lung also depends on the $\dot{V}_A/\dot{Q}$ ratios of the alveoli from which the expired gases originate, and on the solubilities of those gases (29), and is more affected in the high $\dot{V}_A/\dot{Q}$ areas than in the normal areas. This has been shown from the tidal model of ventilation with branching airway structure discussed above (Steady-State Versus Dynamic Models section), where there are a limited number of alveolar units and both fast and slow $\dot{V}_A/\dot{Q}$ alveolar units (85). In diseased lungs, any spread of $\dot{V}_A/\dot{Q}$ ratios is likely to increase ventilation to units with $\dot{V}_A/\dot{Q}$ ratios exceeding unity more than to units with $\dot{V}_A/\dot{Q}$ ratios exceeding 100. The effective ventilation ($\dot{V}_{i, \text{Effective}}$) is affected by mixing in common dead spaces. The composition of the gas in each common dead space depends on the $\dot{V}_A/\dot{Q}$ ratios of the alveolar compartments from which the expired gases originate and the solubilities of the inert gases (29). These differences are demonstrated in Figure 8 for emphysema. The $\dot{V}_{i, \text{Effective}}$ for each inert gas in the high $\dot{V}_A/\dot{Q}$ compartments is more affected than the $\dot{V}_{i, \text{Effective}}$ in the mid $\dot{V}_A/\dot{Q}$ compartments, and as perfusion to each compartment is independent of the effect of common dead space ventilation, the variation in $\dot{V}_{i, \text{Effective}}$ results in variation in the $\dot{V}_A/\dot{Q}$ ratio showing an effect, over the normal $\dot{V}_A/\dot{Q}$ compartment, for both the slow and fast high $\dot{V}_A/\dot{Q}$ compartments as well as a difference between the slow and fast high $\dot{V}_A/\dot{Q}$ compartments themselves.

Schilder et al. (68) showed that the last part of the flow-volume exhalation curve in forced expiration was dependent on the viscosity of the exhaled gas while maximum flow was dependent on the density, with low-density gases such as helium producing the best flows. This is in agreement with the principle that turbulent flow depends on gas density whereas gas viscosity affects laminar flow. Diffusivity also affects the size of the respiratory dead space, as lower density gases have a steeper phase II slope (transition and rapidly rising phase for inert gas partial pressure, from the initial dead-space value to the approximate alveolar value) and shallower phase III slope (equilibration and very slowly rising phase for inert gas partial pressure, from the approximate alveolar value to the final alveolar value) when the exhaled measurement of dead space is undertaken. When gases of similar blood-gas solubility but

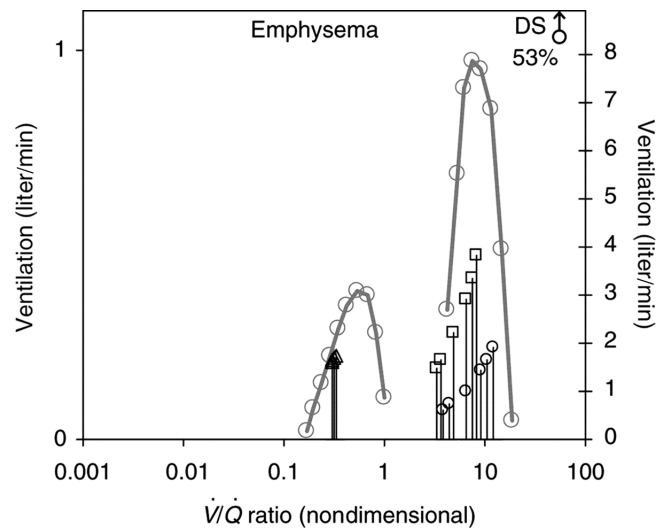


Figure 8 The digitized emphysema MIGET $\dot{V}_A$ distribution (gray, open circles, left ordinate) and the recovered three-compartment $\dot{V}_A$ distributions showing $\dot{V}_{i, \text{Effective}}$ for each inert gas, with solubility increasing from left to right (SF_6 to acetone) demonstrating an increasing effect of the solubility of the inert gas and that the high $\dot{V}_A/\dot{Q}$ areas are affected more than the normal $\dot{V}_A/\dot{Q}$ areas for both slow and fast $\dot{V}_A/\dot{Q}$ compartments. Open triangles, mid $\dot{V}_A/\dot{Q}$ compartment; open squares, fast high $\dot{V}_A/\dot{Q}$ compartment; and open circles, slow high $\dot{V}_A/\dot{Q}$ compartment. DS, dead space. From Yem et al. (85), with permission.

with different gas densities and diffusivity are considered, there have been demonstrated differences in the dead-space volumes measured. He ($S_b = 0.0080$, density 0.165 kg/m^3 , and diffusivity compared to N_2 of 0.7) shows a consistently smaller dead space ($\sim 130 \text{ ml}$ vs. $\sim 150 \text{ ml}$) than does SF_6 ($S_b = 0.0068$, density 6.3 kg/m^3 , and diffusivity compared to N_2 of ~ 0.1) (55). Hlastala et al. (38) showed that for acetylene ($S_b = 0.75$, density 1.1 kg/m^3) and freon 22 ($S_b = 0.67$, density 3.7 kg/m^3), acetylene had a smaller dead space.

Inert gases with high blood solubility, such as acetone ($S_b = 330$) and ethyl alcohol ($S_b \sim 1750$), have virtually no phase I (representing the initial portion of the exhaled breath derived from the anatomical dead space, in which the partial pressure of the inert gas is normally zero). This is because these gases are excreted into the airways rather than by alveolar excretion, so there is little or no dead space.

It has been assumed that the dead space is distributed in proportion to the alveolar ventilation (29, 71), which will also be affected by the use of gases of different solubilities to determine the dead space. This difference in measured dead space due to gases of different solubilities does not appear to have been used to investigate pathological lung states.

Anesthetic uptake and elimination models

So far, an ODE model has been developed that incorporates a parallel array of tissue-capillary elements of known volume, perfusion, and tissue-blood partition coefficient. The arterial partial pressure can be set to a chosen value but remains

constant during the wash-in or washout. The flow-weighted average of the partial pressures of the venous effluents of each compartment yields the mixed venous partial pressure. However, according to this model, any difference in the wash-in and washout kinetics between different inert gases into a particular tissue compartment depends solely the tissue-blood partition coefficient. Consequently, Figure 5 demonstrates that the wash-in and washout kinetics for the VRG are very similar for desflurane and ether, despite the former producing anesthesia and awakening that is an order of magnitude more rapid than the latter.

This surprising finding is the result of the oversimplification of the model, which so far has assumed that the arterial partial pressure can be switched instantly to a constant value of one's choosing. For example, at the end of anesthesia, it is assumed that the arterial partial pressure of the inert gas can be switched instantly to zero simply by removing any vapor from the inspired air. This is an unrealistic assumption because even when the inspired vapor concentration is switched off, vapor recirculating in the mixed venous blood will feed into the arterial blood to various degrees depending on the $\dot{V}_A/\dot{Q}$ ratio of the lung and the blood-gas solubility, S_b . This phenomenon is the basis of the MIGET technique (discussed in detail elsewhere in *Comprehensive Physiology*) and can be quantified for the elimination phase (for the steady state) by the following mass-balance equations:

The volume of vapor returning to the lung in the mixed venous blood per unit time is given by:

$$S_b \dot{Q} P_v$$

The volume of vapor leaving the lung in the arterial blood per unit time is given by:

$$S_b \dot{Q} P_a$$

The difference in these rates equals the rate of removal of vapor by ventilation of the lung plus the rate of accumulation of vapor in the lung. Hence,

$$S_b \dot{Q} P_v - S_b \dot{Q} P_a = \dot{V}_A \frac{P_a}{P_B} + \frac{V_L}{P_B} \frac{dP_a}{dt} \quad (6)$$

where V_L is the lung volume and P_B is barometric pressure, which for simplicity is assumed to be equal to 1.

Just considering the steady state where $dP_a/dt = 0$, Eq. (6) can be simplified to the following:

$$P_a = P_v \left[\frac{1}{1 + \frac{1}{S_b} \frac{\dot{V}_A}{\dot{Q}}} \right] \quad (7)$$

From Eq. (7), it is apparent that the (steady state) arterial partial pressure is coupled to, and therefore governed by, the mixed venous partial pressure. The coupling factor in the

square brackets is seen to depend on the blood-gas solubility and the mean $\dot{V}/\dot{Q}$ ratio. For very small and diminishing values of S_b , the denominator of the coupling factor tends to infinity and so the factor itself tends to zero. In these circumstances, the arterial partial pressure is independent of the mixed venous partial pressure and there is no feed-through of gas from the mixed venous blood to the arterial blood. The same is true for infinite values of $\dot{V}/\dot{Q}$, so the effect of low blood-gas solubility is equivalent to extreme hyperventilation. Conversely, both increasing S_b and decreasing $\dot{V}/\dot{Q}$ increase the coupling between P_a and P_v and hence increase the feed-through of gas from recirculating mixed venous blood into the arterial circulation. So while in the constant arterial pressure model described in the Models and Algebraic Analysis section, it is only the tissue-blood partition coefficient that determines the kinetics of tissue uptake and elimination, in the more realistic "inconstant arterial pressure" model seen in real-life elimination, it is both the tissue-blood partition coefficient and the blood-gas solubility that are important.

Such realistic models, incorporating the interplay between recirculating mixed venous blood and the arterial circulation, are of interest to anesthesiologists and physiologists. Numerical analyses and computer simulations soon followed Kety's early analytic multicompartment description. Notable models were those of Mapleson, Zwart, and Cowles (17, 53, 89). Based on these analyses, Philip produced the computer program GasMan[®] (Med Man simulations, Inc., PO Box 67160, Chestnut Hill, MA 02467, USA), a four-compartment model designed as a teaching aid, which has been widely used in subsequent investigations (60, 61). GasMan[®] and other simulations include the effects that differences in $\dot{V}/\dot{Q}$ ratio and S_b values have on the arterial and tissue partial pressures when step changes are made in the inspired, rather than the alveolar partial pressure, and incorporate this in a more comprehensive model of wash-in and washout kinetics.

The interested reader can undertake this analysis numerically for the non-steady-state condition with reasonable accuracy by using a simple spreadsheet. First, Eq. (6) is solved at $t = 0$ for dP_a/dt . This latter term is then integrated over a small time step to give a value for P_a at time $t + \delta t$, and this iteration is repeated as t increases. Simultaneously, Eq. (2) is solved at $t = 0$ for dP_v/dt for each compartment, and successive values of P_v are calculated by integration as before. Finally, the simultaneous values of mixed venous partial pressure, required in Eq. (6), are calculated at each time point from Eq. (5).

A non-steady-state analysis of this kind is shown in Figure 9.

This analysis yields markedly different results from those shown in Figure 5, and the differences between the highly blood-soluble ether and highly insoluble desflurane are now clearly seen. At 10 min after cessation of inhalation of these two vapors, the partial pressures in the VRG are seen to be approximately 7% (desflurane) and 53% (ether) of the pre-washout value. The difference between the simple model shown in Figure 5 and the realistic integrative model

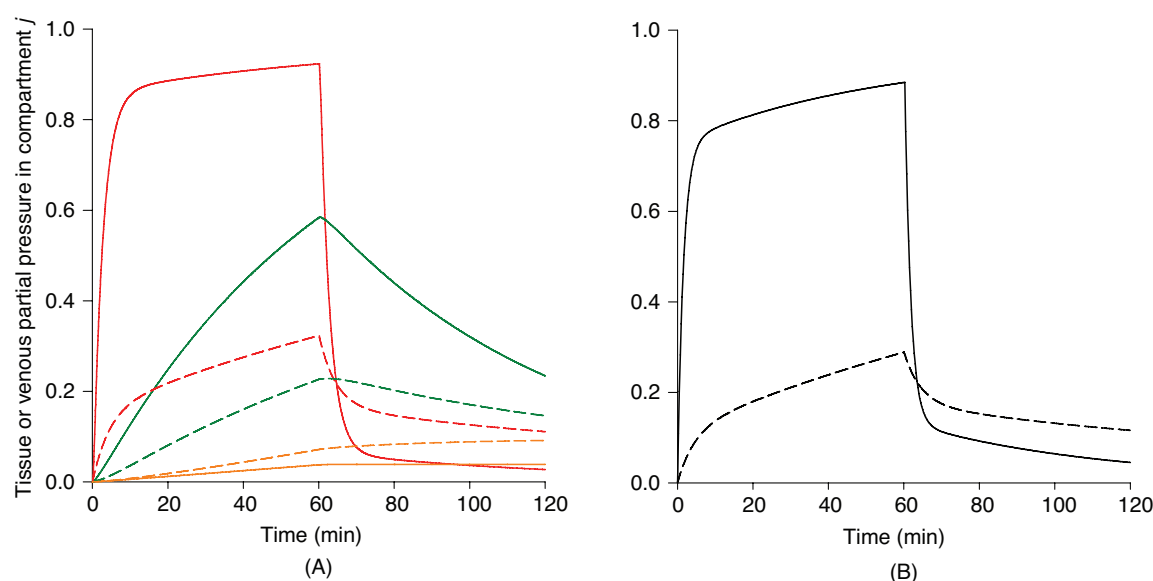


Figure 9 (A) Compartmental venous and tissue partial pressure versus time after imposition of a constant inspired partial pressure of 1.0. At 60 min, the inspired partial pressure is set to zero. Red lines, vessel-rich group; green lines, muscle group; and orange lines, fat group. Solid lines represent the anesthetic vapor desflurane, dotted lines represent ether. (B) The mixed venous partial pressures for the same simulation.

shown in Figure 9 is exaggerated by agents such as acetone. Table 2 shows the λ values of acetone to be very low for all tissue groups, even lower than those for desflurane. Based on this alone, the simple model would predict rapid washout of this vapor. However, the S_b for acetone is very high (much higher even than ether), and so even though the capacity of the tissue reservoir for acetone is small, the large S_b would result in minimal elimination of vapor via the alveoli. The arterial partial pressure would therefore decay only very slowly, and this would result in much slower washout than predicted by the simple model.

Context sensitivity

The decrements in VRG partial pressure to 7% and 53% as quoted above depend not just on the properties of the vapor but also on the duration of exposure to it. Eger and Schafer (21), using GasMan[®], showed that increasing the duration of anesthesia increases the time required to wash out the VRG to a given decrement in a nonlinear manner, and this effect was exaggerated in more blood-soluble agents. The decrement times of desflurane had little exposure time sensitivity when considering decrements down to partial pressures of between 8% and 20% of pre-washout values. They showed that decrement time differences between agents were particularly exaggerated if one chooses to compare times taken for the VRG pressure to fall to very low levels, say below 5%, but this may have little clinical relevance since once the VRG is washed out to 5% of its initial (anesthetically effective) value, the clinical effect would probably be small and so there is little interest in the relative speed of washout from this point on.

Context sensitivity is a function of both the tissue-blood partition coefficient and the blood solubility. The tissues act as a reservoir for dissolved vapor, and an increase in tissue solubility results in increased capacitance. However, the blood solubility determines the conductance, that is, the capacity of blood to carry vapor to and from the tissue. As discussed in the Models and Algebraic Analysis section, the capacitance:conductance ratio determines the time constant of wash-in and washout. As the duration of exposure increases, the tissue compartments become more saturated. It can be seen in Figure 9 that this process is far from complete in all compartments except the VRG even after 1 h. The fat compartment particularly has the capacity to store more and more vapor as time progresses. However, regardless of the size of the body tissue reservoir, if during the washout period returning venous blood is completely cleared of vapor as it passes through the lungs (i.e., if the blood-gas solubility is zero), there will be no feed-through of vapor into the arterial blood and so the size of the body tissue reservoir is irrelevant. For intermediate values of blood solubility, both λ and S_b together determine washout times and exposure time sensitivity. It is often mistakenly assumed that since the rate of rise of arterial partial pressure of an anesthetic agent is determined solely by its blood-gas solubility, then the rate of fall of partial pressure at the end of exposure will be similarly determined solely by its blood-gas solubility. It is often assumed that the washout is merely a symmetrical reflection of the wash in. While this is almost true for very brief exposures (during which the tissue capacitance has not been filled), it is not so for longer exposures during which tissues (notably fat) will have accumulated a reservoir of agent. This explains why the wash-in kinetics of desflurane and N₂O ($S_b = 0.42$

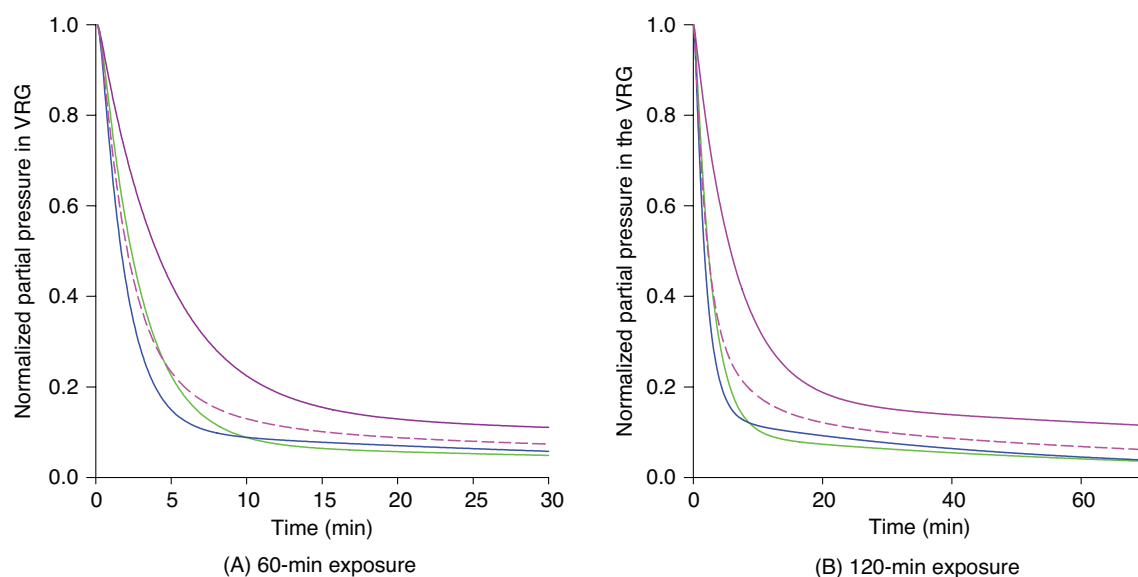


Figure 10 Washout of desflurane (green), N₂O (blue), and isoflurane (purple) and a 30:70 MAC ratio mixture of isoflurane and N₂O (pink dashed) from the VRG after 60 and 120 min of exposure at 1.0 MAC inspired throughout. Partial pressures are normalized, that is, expressed as a fraction of the value at $t = 0$.

and 0.47, respectively) are virtually identical, yet the washout of N₂O is more rapid than desflurane since the tissue-blood partition coefficients are greater for desflurane in all tissue groups (most notably in fat, see Table 2). In practice, both of these agents have relatively rapid washouts, which are quite context insensitive, when compared to agents with higher S_b and λ values such as halothane.

Gas mixtures

As shown in Figure 9, the partial pressure of desflurane in the VRG rises and falls rapidly on induction of, and emergence from, anesthesia. The less modern and more blood-soluble isoflurane has more sluggish kinetics, and for prolonged anesthesia, washout of the VRG to say 10% of the clinically effective partial pressure is prolonged relative to desflurane.

The anesthetic potency of an agent is given by its minimal alveolar concentration (MAC), which is ED₅₀, that is, the minimal partial pressure required to prevent gross movement to painful stimuli in 50% of subjects. In practice, the

MAC or ED₅₀ of agents is additive so that if two different agents are supplied at 0.5 MAC each, the resulting mixture will have a MAC of 1.0. Quoting anesthetic partial pressures as fractions of an MAC allows agents of different potencies to be directly compared. If a mixture of N₂O providing 0.7 MAC and isoflurane providing 0.3 MAC is administered, the resultant washout of the mixture, expressed in fractions of a summated MAC value, can be predicted using the model described. Figure 10 shows the results of this washout for exposures of 60 and 120 min before washout commences.

The times taken for the VRG partial pressures to fall to various arbitrary thresholds is shown in Table 3. For exposure times up to 2 h, the times for partial pressures (expressed as MAC fractions) to fall to 15% of the initial value are similar for desflurane and the isoflurane/N₂O mixture.

Airway obstruction during inhalation of anesthesia

When emergency anesthesia is required and there is acute partial obstruction of the upper airway due to, for example,

Table 3 Times in Minutes for VRG Partial Pressure to Fall to 0.2, 0.15, and 0.10 MAC for Isoflurane, Desflurane, and a 30:70 MAC Ratio Mixture of Isoflurane and N₂O

Decrement to (MAC Fraction)	Exposure time	Isoflurane		Desflurane		Isoflurane/N ₂ O	
		60 min	120 min	60 min	120 min	60 min	120 min
0.20		11.1	18.0	5.5	5.7	5.8	8.4
0.15		15.6	31.3	6.7	7.2	8.2	13.4
0.10		35.0	75.0	8.8	10.7	15.0	29.0

Data are shown for exposure times of 60 and 120 min.

trauma, edema, tonsillar infection, or hemorrhage, the induction of anesthesia by inhalation (rather than by intravenous agents) is considered advisable. The reason for this is because if complete airway obstruction supervenes and ventilation were impossible during the attempt, no further anesthetic vapor would be inhaled and the patient would resume consciousness and spontaneous ventilation. At first sight, the mechanism for this spontaneous recovery may not be obvious, because whilst obstruction and apnea would prevent further loading of anesthetic vapour it would also prevent washout. Why then should the patient awaken? The reason is that during airway obstruction and apnea, while vapor is not eliminated from the body, it is redistributed from the brain to other compartments (i.e., elsewhere in the VRG, MG, and FG). Under these circumstances, the mean $\dot{V}/\dot{Q}$ ratio is zero and so as can be seen from Eq. (7), the value of the blood solubility, S_b , becomes irrelevant and the arterial partial pressure tends to the mixed venous value. In turn, the mixed venous partial pressure is determined by the upstream tissue partial pressure, and this is flow weighted toward the nonbrain “sink” compartments since these receive four fifths of the cardiac output.

From Figure 2, we see that the tissue partial pressure of the sink compartments will be low if they have a large volume and high tissue-blood partition coefficient. As the $\dot{V}/\dot{Q}$ ratio is nil, this low partial pressure will feed through unchanged to the arterial circulation. Under these circumstances, therefore, the redistribution kinetics are determined by the tissue-blood partition coefficient of the agent concerned.

Talbot and colleagues studied this mechanism in volunteers who underwent simulated inhalational induction and simulated airway obstruction (73). The experiments were designed to deliver partial pressures at one-tenth of the values that might be used clinically for the agents sevoflurane (low blood and tissue solubility) and halothane (high blood and tissue solubility). Since the system is linear, the kinetics are not affected by the absolute partial pressures and so scaling down the partial pressure to one-tenth allowed the subjects to be clinically unaffected by the inhalation. At a point during the inhalation when the alveolar partial pressure reached 0.2 MAC (equivalent to 2 MAC in the upscaled clinical equivalent and when consciousness and airway patency might be lost), simulated airway obstruction was imposed. This was achieved by diverting the breathing system to force the subject to rebreathe into a small bag. This produced an effective alveolar ventilation rate of zero yet still allowed end-tidal gas sampling by the mass spectrometer. Sample data from two experimental runs in one volunteer are shown in Figure 11. Halothane concentration in the blood/VRG falls more rapidly than sevoflurane. When fitted to a monoexponential decay function, the time constants for halothane and sevoflurane from the pooled data were 22.1 and 62.1 s, respectively, demonstrating that in the unusual circumstances where alveolar ventilation rate is zero, it is the tissue-blood partition coefficient (rather than the blood-gas solubility) that determines the rate of redistribution of volatile agents from the VRG.

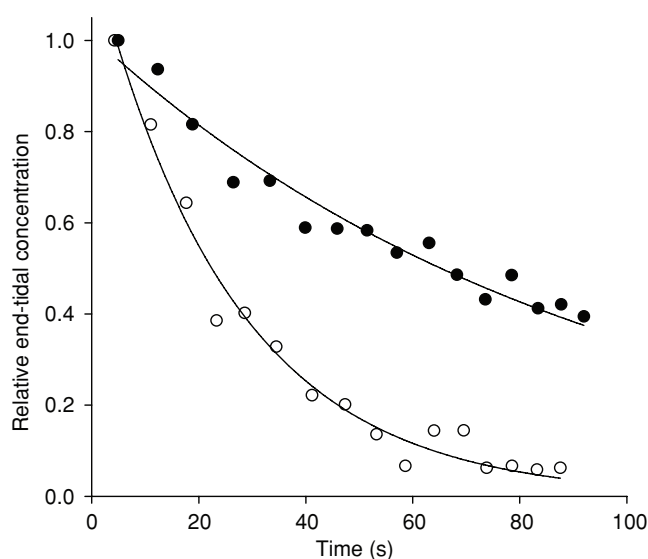


Figure 11 Sample data from two experimental runs in one volunteer showing the decline in end-tidal sevoflurane and halothane concentrations during rebreathing from a 1-liter bag. Filled symbols, sevoflurane; open symbols, halothane. Solid lines show monoexponential model fits ($y = a e^{-t/\tau}$) for each set of data points. Concentrations are expressed relative to the first end-tidal value recorded during rebreathing.

Cavity Gas

Cavity gas mixtures

Some inert gases are used in medical therapy as splints to hold tissues in place. The main role for this is in eye surgery where sulphahexafluoride (SF_6) and perfluoropropane (or octafluoropropane) (C_3F_8) are commonly used. Xenon (Xe) and perfluoroethane (or hexafluoroethane) (C_2F_6) are less commonly used. The effect on the gas bubble of the inspired gases, such as oxygen, N_2O and air is of interest, as is the expansion of the bubble with increased altitude consistent with Boyle's law (30) (see Table 4).

If 100% of the inert gas were used, when the patient subsequently breathed air, there would be an expansion of the intraocular bubble as nitrogen diffused into it. Therefore, the inert gas is usually diluted with air to minimize expansion of the intraocular bubble on resumption of air breathing. The mixtures with air used clinically are individually 20% SF_6 , 16% C_2F_6 , or 12% C_3F_8 (14). If, however, the patient subsequently has a general anesthetic that includes N_2O (or xenon, another anesthetic gas at a high concentration), then N_2O will enter the inert gas bubble more quickly than nitrogen will diffuse out and the bubble will therefore expand and compress the retinal vessels. Expansion of gas bubbles with a xenon-oxygen mixture has been shown to be less than that for a N_2O - O_2 mixture (10,33,67). Patients who have received intraocular gas injections are advised not to fly while the gas remains because the reduced ambient pressure during flight will expand the gas bubble. Because of their low blood-gas solubilities, C_3F_8 has a volume half-life of approximately 30 days, C_2F_6 approximately 15 days, with SF_6 approximately

Table 4 The blood-gas solubilities of the inert gases used for intraocular gas splinting of tissues, together with nitrogen, oxygen, and nitrous oxide

Gas	O ₂	N ₂	N ₂ O	SF ₆	C ₃ F ₈	C ₂ F ₆	Xe
S _b , ml/ml/atm	0.0227 ^a	0.0146 ^a	0.468 ^b	0.008 ^c	0.00125	0.00277	0.115 ^d

Data from Chortkoff et al. (15).

^aBurkard and Van Liew (13).

^bFiserova-Bergerova and Diaz (26).

^cAnderson and Hlastala (4).

^dGoto et al. (32).

7 days (14), so the risks associated with N₂O anesthesia and air-flight may persist for some time.

Expansion of N₂O within the middle ear may occur when there is a blockage of the Eustachian tube because N₂O will diffuse into the air cavity at a greater rate than nitrogen can diffuse out. By the same mechanism, volume shrinkage occurs postoperatively if N₂O has been used during surgery and tympanic membrane grafts may therefore be displaced.

If the Eustachian tube is blocked in an individual subjected to a low barometric pressure during air-flight, there is a risk of perforation of the eardrum due to the expansion of middle ear gas even when that gas is air.

Nitrous oxide will diffuse into a pneumothorax or into luminal bowel gas (which accumulates in the presence of bowel obstruction) and will therefore expand the pneumothorax or distend the bowel yet further. Nitrous oxide anesthesia is relatively contraindicated in these circumstances.

Alveolar gas resorption

A special situation for gas absorption occurs in the lung distal to an airway blockage due to a mucus plug or other obstruction. When the subject is breathing air, and the lung contains air, then the alveolar volume distal to the blockage diminishes relatively slowly as oxygen is absorbed into the pulmonary capillary blood. Nitrogen in the gas pocket is then concentrated and slowly resorbed to maintain partial pressure equilibrium. The pocket will collapse to full alveolar atelectasis over approximately 350 to 525 min (40). As the solubility of the gas mixture increases, so the time to collapse is shortened. Conversely, with less soluble gases such as He, the time to collapse takes longer. Inhaling mixtures of O₂ and N₂O causes interesting effects on the rate of atelectasis of obstructed airspaces.

If 100% O₂ is inhaled when a small airway becomes obstructed, the rate of atelectasis is increased compared to air breathing (collapse occurs within 6–7 min). This occurs because the trapped gas is mostly oxygen, which diffuses out of the alveolus into the pulmonary capillary at a roughly constant (perfusion limited) rate. The space contains no nitrogen to delay its complete collapse.

If a 50:50 N₂O/O₂ mixture is being inhaled prior to airway obstruction, there is a further increase in the rate of collapse that occurs within 4 to 5 min. This occurs because, as before,

oxygen diffuses from alveolus to pulmonary capillary. As the uptake is perfusion limited, the rate is the same rate as was the case for 100% oxygen. However, under these circumstances, oxygen uptake results in a concentrating effect on the trapped N₂O and so its partial pressure now exceeds that of the mixed venous blood with which it was previously in equilibrium. Because of its high solubility, it too can rapidly diffuse out of the alveolus alongside the oxygen.

As the inspired N₂O fraction (F_{IN₂O}) increases to 80%, the rate of collapse *decreases* somewhat (15–20 min to collapse) (Fig. 12. This can be explained by the fact that oxygen now

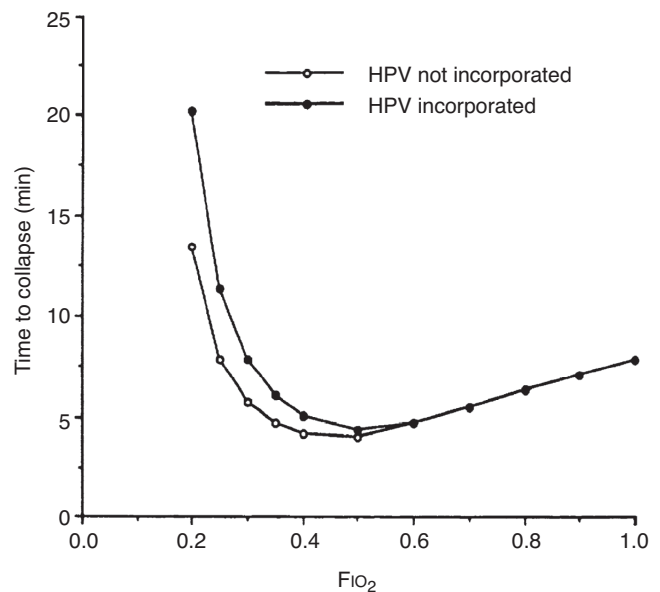


Figure 12 When ventilation to an area of lung ceases, lung collapses at a rate dependent on the inspired gas composition. Shown is the time to collapse found when breathing various mixtures of O₂ and N₂O was modeled. Results are given for when hypoxic pulmonary vasoconstriction (HPV) was, and was not, incorporated into the model. As F_{IO₂} decreases from 1.0, time for collapse decreases to a minimum at F_{IO₂} = 0.5. Further decreases in F_{IO₂} result in increase in time to collapse. When HPV was incorporated into the model, time to collapse was approximately the same with F_{IO₂} = 1.0 as with F_{IO₂} = 0.3. When HPV was not incorporated, collapse took about the same time with F_{IO₂} = 1.0 as with F_{IO₂} = 0.25. Inclusion of HPV in the model resulted in prolongation of time to collapse with increasing effect as F_{IO₂} decreased. Maximal prolongation was at F_{IO₂} = 0.2 (lowest F_{IO₂} considered) when time to collapse was 50% longer with than without HPV incorporation. From Joyce et al. (40), with permission.

occupies only a small fraction of the gas space and so its disappearance has only a modest concentrating effect on the N_2O , which moderates the rate of diffusion of the latter from the space (40).

Figure 12 demonstrates that the usual range of concentrations of N_2O used in anesthesia (50%-70%) is in the range that induces lung atelectasis fastest following any subsequent airway blockage.

When 100% oxygen is breathed for periods of 3 min or more before subsequent administration of N_2/O_2 or N_2O/O_2 mixtures (pre-oxygenation), there is a marked decrease in time to collapse (4-8 times faster) for both mixtures at all inspired oxygen fractions (F_{IO_2}), with N_2O/O_2 mixtures slightly faster than N_2/O_2 , though not clinically significant (41). This effect is different from the decrease in functional residual capacity (FRC), which occurs with the induction of general anesthesia due to the effect of relaxation of muscle tone particularly in the diaphragm and which also causes atelectasis (75). Another factor contributing to perioperative atelectasis is the effect of breathing high concentrations of oxygen during the recovery phase from anesthesia when there is a reduced FRC and hypoventilation due to a number of factors such as pain, decreased muscle tone from residual muscle paralysis and anesthesia, and lowered respiratory drive without the usual intermittent large tidal breaths that normally occur. Baker and colleagues have shown that there is a prolonged decrease of up to 10% in both residual volume (RV) and FRC following a short period of 5 min breathing high inspired oxygen levels at reduced FRC (7, 8). The magnitude of this effect is dependent on the F_{IO_2} and disappears only for an F_{IO_2} of less than 0.25 (7, 8). This effect on lung volumes may also be important when oxygen breathing occurs both in aviation with high g-forces and in diving with reduction in FRC due to hydrostatic pressure.

Measurement

Measurement of the uptake, disposition, and distribution of inert gases within a perfused organ can be used to quantify its perfusion. The underlying physical principles of these techniques are grounded in Fick's principle, which itself is an outworking of the principle of the conservation of mass.

Measurement of peripheral organ perfusion

Different techniques are suitable for different circumstances. Some depend on the assumption of a specific kinetic model and complete equilibration between blood and tissue; others make no such assumptions. Three basic techniques are described below.

Measurement of prolonged inert gas uptake

This technique has become known as the Kety-Schmidt technique, after the investigators who described the technique to

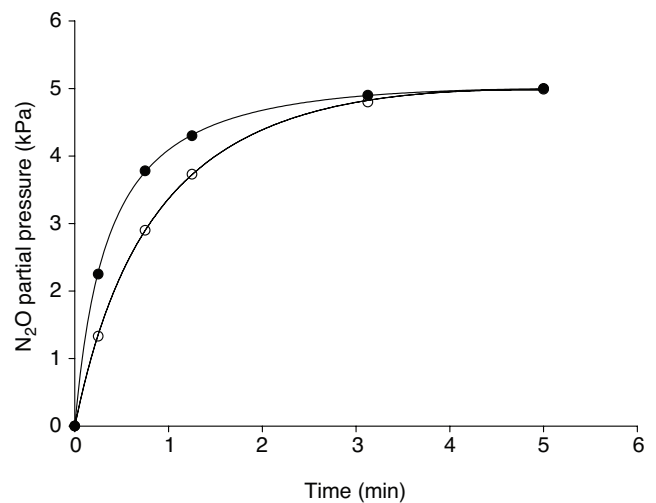


Figure 13 The Kety-Schmidt method of prolonged N_2O wash-in to the brain via inhalation. Closed circles show arterial sampling; open circles show cerebral venous sampling.

measure cerebral blood flow by using N_2O uptake (46). Nitrous oxide is inhaled at a constant fraction for a period of 15 min during which intermittent arterial and cerebral venous blood sampling is undertaken. Because of its low blood-gas solubility, the alveolar and arterial partial pressures rise quickly toward the inspired value. The brain, in turn, equilibrates with the arterial partial pressure at a rate determined by the blood flow per unit volume and the tissue-blood partition coefficient. The equilibration should not be so fast that intermittent sampling is unfeasible, nor so slow that the measurement process is excessively long, and equilibration never achieved. Since cerebral blood flow is high, the relatively low blood-tissue partition coefficient of N_2O ($\lambda = 0.8$) provides a suitable equilibration time of 5 to 10 min. Figure 13 shows a representation of such data.

The mass balance equation can be derived as follows:

The rate at which the inert gas is being delivered to the organ is given by:

$$\dot{Q} S_b P_a(t)$$

where $\dot{Q}$ is the arterial blood flow and P_a the arterial partial pressure of inert gas.

The rate at which the inert gas leaves the organ is given by:

$$\dot{Q} S_b P_v(t).$$

The difference between these rates equals the rate of accumulation or uptake by the organ, which when integrated yields the mass of inert gas within the organ at equilibrium,

that is, $V_{ti} S_{ti} P_v(t_{\infty})$. Hence:

$$\int_0^{\infty} (\dot{Q} S_b P_a(t) - \dot{Q} S_b P_v(t)) dt = V_{ti} S_{ti} P_v(t_{\infty}) \quad (8)$$

Since the perfusion and blood solubility are assumed to be constant, Eq. (8) reduces to:

$$\dot{Q} S_b \int_0^{\infty} (P_a(t) - P_v(t)) dt = V_{ti} S_{ti} P_v(t_{\infty}) \quad (9)$$

The integral term in Eq. (9) is the area between the arterial and venous curves in Figure 13. Therefore, the perfusion per unit of tissue volume is given by:

$$\frac{\dot{Q}}{V_{ti}} = \frac{\lambda P_v(t_{\infty})}{\text{Area between curves}} \quad (10)$$

The Kety-Schmidt technique has been used with N_2O , Ar, Ne, H_2 , ^{85}Kr , and ^{133}Xe . Organs with high specific blood flows are best measured with gases of lower partition coefficient, and vice versa, such that the conductance: capacitance ratio is manipulated to give equilibration within 1 to 5 min. If the tracer is too soluble or if the blood flow too low, equilibration will be prolonged and the area between the curves will be only a small fraction of the total measured areas. Any error in the measured area under the curves (AUC) will feed through into the area between the curves (ABC), and this will become increasingly significant as the $ABC \ll AUC$.

The technique measures the mean flow over the whole measurement period and is therefore unsuitable for dynamically changing flows. It is also assumed that the tissue is homogeneous with respect to the partition coefficient. When this is not the case, the returned perfusion value is the “as if” equivalent perfusion of a uniform tissue with a single partition coefficient.

Measurement by arterial bolus injection of gas in solution

If a bolus of a radioactive tracer gas (in solution) is injected intra-arterially, then the rate of disappearance of the tracer from the organ gives an indication of the perfusion. This technique offers a potential advantage over the Kety-Schmidt technique because by measuring the tracer activity with an external counter, the need for rapid venous sampling is obviated. If the tracer gas has low blood-gas solubility, then it will be largely cleared by a single pass through the lungs and so recirculation to the tissue will not occur. A variation of this technique is to inject the tracer directly into the tissue rather than intra-arterially, and this was first described by Kety in 1949 (43). Kety’s analysis was based on the assumption that the tracer, once injected into the tissue, achieved diffusive equilibrium instantly and that the incoming blood simply di-

luted the tracer exponentially, as one might expect if fresh water flowed into a bucket of well-stirred inky water and an exponentially diluted effluent flowed out of the overflow at the same rate as the inflow. This led to attempts to fit the observed data to an exponential model of the following form:

$$\frac{dm(t)}{dt} = -\frac{\dot{Q}}{\lambda V_{ti}} m(t) \quad (11)$$

where $m(t)$ is the externally measured radiation count, denoting the quantity of tracer present.

This describes a simple exponential decay whose rate constant is $-\dot{Q}/\lambda V_{ti}$. However, the problem with this approach is that experimental data do not support the model shown in Eq. (11), that is, they seldom resemble a monoexponential function. Ingvar and Lassen (39) considered that the failure of the data to fit a monoexponential model was due to the fact that the organ (in their case, the brain) was functionally multi-compartmental, as if fresh water were now flowing in parallel into two separate and dissimilar buckets of inky water whose overflows were conjoined. They used an exponential stripping technique to separate out two functional compartments. Zierler warns against inferring anything anatomical or even functional from the ability to strip experimental data into discrete compartments, especially since there is no real basis for the assumption that the function is exponential at all (87).

Instead, Meier and Zierler propose a different model (54), one in which no assumption about the nature of the indicator kinetics is required. Moreover, no assumption about equilibration between blood and tissue is necessary. In this analysis, the tracer is considered to be “distributed” within the organ rather than classically “diluted” within it. To understand this, we need to consider the concentration-time relationship in the common venous effluent of a rapidly and intra-arterially injected tracer. Such an example is shown in Figure 14.

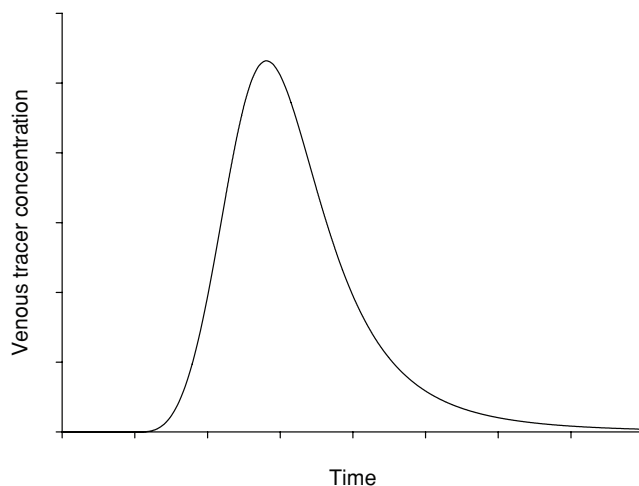


Figure 14 Venous concentration-time profile following rapid intra-arterial injection of a bolus of tracer.

Figure 14 shows, not surprisingly, that there is a delay in the appearance of tracer in the venous effluent after bolus intra-arterial injection. Since the arterial injection is rapid (defined later), the concentration of tracer in the myriad of tissue arterioles and capillaries will be equal since they just represent an array of parallel run-offs from the injection site. After a brief lag, the tracer, which has traversed the shortest capillaries, or more specifically those capillaries with the shortest transit times, begins to enter the common venous effluent and so the venous concentration begins to rise. Since the number of capillaries with such short transit times in the population is small, the contribution of the tracer at this point to the overall venous concentration is small. The peak venous concentration corresponds to the time when those capillaries with the commonest transit times (i.e., the transit time “mode”) all discharge their tracer together into the venous outflow. After this point, venous concentration falls because only the few capillaries whose transit times are long are continuing to add the tracer to the venous blood; those with shorter transit times are by now contributing tracer-free blood to the flow.

From this description, it can be appreciated that the concentration-time profile represents the frequency distribution (or probability density) of transit times within the capillary bed. In fact, when appropriately scaled, the two functions are identical.

Let us call this frequency distribution $h(t)$. This distribution also describes the rate of loss of tracer from the system $\dot{m}(t)$ at any time t since

$$\dot{m}(t) = m_0 h(t) \quad (12)$$

where m_0 is the initial amount of tracer injected.

Since the rate of loss of tracer at time t is also given by the product of the flow and the concentration, Eq. (12) can be expanded to

$$\dot{m}(t) = m_0 h(t) = \dot{Q} c(t)$$

Hence,

$$h(t) = \frac{\dot{Q}}{m_0} c(t) \quad (13)$$

Here, we see that the venous concentration-time relationship is identical to the transit time probability density function, albeit scaled by the factor $\dot{Q}/m_0$.

The importance of the mean transit time

Considering only the capillaries whose transit time is t , we can state that the portion of flow that traverses these capillaries is $\dot{Q} h(t)$. The volume of these capillaries, δV , is the flow multiplied by the transit time, hence:

$$\delta V = \dot{Q} h(t)t$$

If we summate all such volume elements for all values of t , we get the total capillary volume V as follows:

$$V = \int_0^\infty \dot{Q} h(t)t dt = \dot{Q} \int_0^\infty h(t)t dt \quad (14)$$

Since $h(t)$ is the frequency function of transit times, the integral term on the far right in Eq. (14) will be recognized as simply the mean transit time, $\bar{T}$, and so the total flow per unit total volume of the system is given by:

$$\frac{\dot{Q}}{V} = \frac{1}{\bar{T}} \quad (15)$$

It is now clear that the specific blood flow can be determined simply by measuring the mean transit time.

Determining the mean transit time

Figure 15 shows two further functions that are related to the probability density function $h(t)$. While $h(t)$ is the frequency distribution, its integral, $H(t)$, is the cumulative frequency distribution. So if $h(t)$ represents the rate of removal of the tracer as a fraction of the initial tracer mass, Eq. (12), then $H(t)$ represents the cumulative amount of tracer removed from the system as a fraction of the initial amount injected. The amount remaining in the system, $m(t)$, which is measured by an external radiation counter, therefore, is given by:

$$m(t) = m_0(1 - H(t)) \quad (16)$$

If we now examine the area under the $m(t)$ curve (Fig. 15C), we get, by integration of Eq. (16), the following:

$$\int_0^\infty m(t)dt = m_0 \int_0^\infty (1 - H(t))dt \quad (17)$$

It can be shown, although it is not proven here, that for any frequency distribution $h(t)$, the term $\int_0^\infty (1 - H(t))dt$ (i.e., 1 – the double integral of the original distribution) equals the mean of the distribution (in this case, $\bar{T}$). Therefore, from Eqs. (15) and (17), we have:

$$\frac{\dot{Q}}{V} = \frac{1}{\bar{T}} = \frac{m_0}{\int_0^\infty m(t)dt} \quad (18)$$

From Eq. (18), it is clear that the blood flow per unit volume can be determined from the measurement of the initial quantity of tracer detected externally divided by the area under the external activity-time curve. This is equivalent to the maximum value of Figure 15A divided by the area under it.

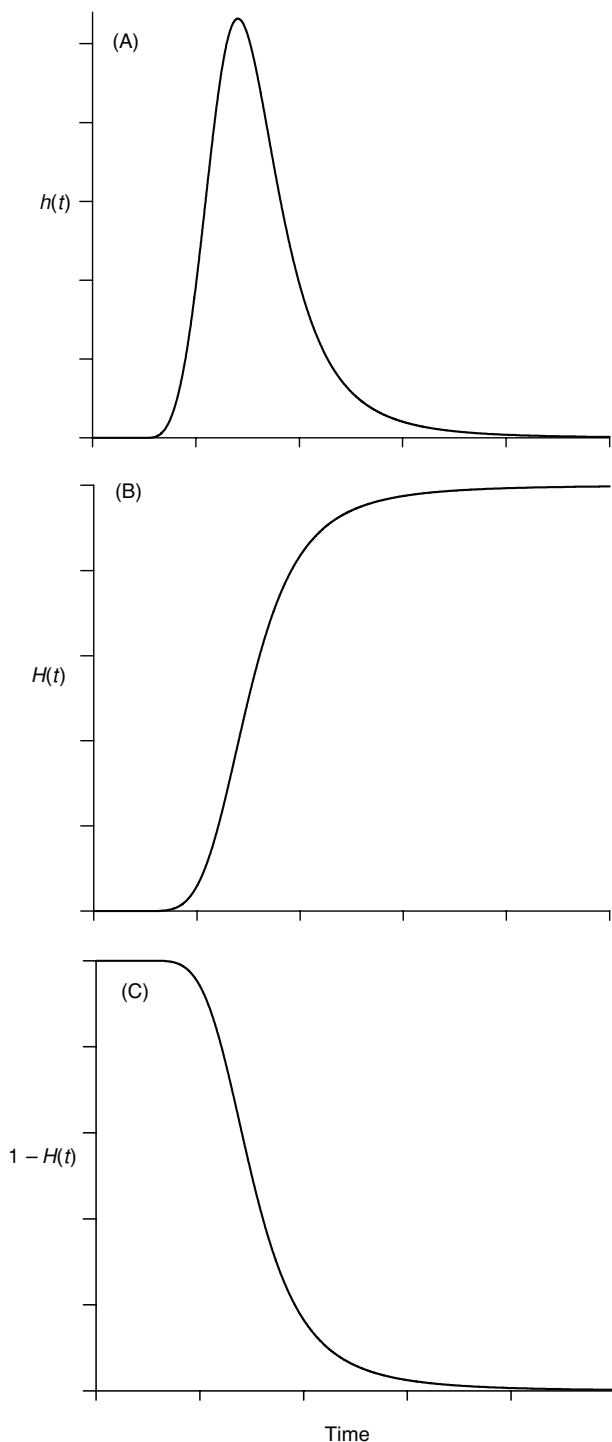


Figure 15 Transit time probability function, $h(t)$ (upper trace), cumulative probability, $H(t)$ (middle trace), and $1 - H(t)$ (lower trace).

We can only take the initial amount of tracer, m_0 , to be equal to the maximum value of the external counts (i.e., the height of Figure 15A) if the bolus injection is such that all the tracer has entered the tissue before any of the tracer has left it, and so this forms the basis of a working definition of what comprises a “bolus” injection.

What is meant by the term “volume”?

In the above analyses, perfusion is quoted as flow per unit volume. But what exactly does this volume term comprise? If the tracer were firmly attached to the red cell membrane, and so were contained within the capillary bed, then this volume term would equal the volume of the capillary bed. However, for an inert gas that is dissolved initially in the plasma, yet is free to diffuse into the interstitial space and the tissue itself, then the “volume” includes the capillary volume and at least some fraction of the interstitial and tissue volumes. The size of this fraction depends on the degree of equilibration that cannot be assumed to be complete. However, Meier and Zierler (54) showed that, in the model described, the effective volume of distribution of a bolus injection of inert gas and steady-state infusion appeared the same. Strictly speaking, the mean transit time of an inert gas particle injected as a bolus is unaffected by the degree of equilibration. This can be appreciated by the inspection of Figure 16.

If equilibration were rapid and complete following a bolus injection, then a particle of tracer would leave the capillary and enter its volume of distribution, which would equal the total tissue volume. It would similarly rapidly leave the tissue to rejoin the venous capillary and its dwell time in the tissue compartment would delay its mean transit time. If diffusion were insufficiently rapid to allow complete equilibration of the tissue, then the volume of distribution of a particle of tracer would be smaller than the tissue volume. However, the dwell time of a tracer particle diffusing in and out of a smaller volume at a slower rate would be the same as the case in which complete equilibration occurred. Given that the mean transit time is therefore unaffected by the completeness of equilibration, the measured flow per unit volume (which equals $1/\bar{T}$) is similarly unaffected. The only correction that is required is that the tissue volume needs to be multiplied by the tissue-blood partition coefficient so that the volume term

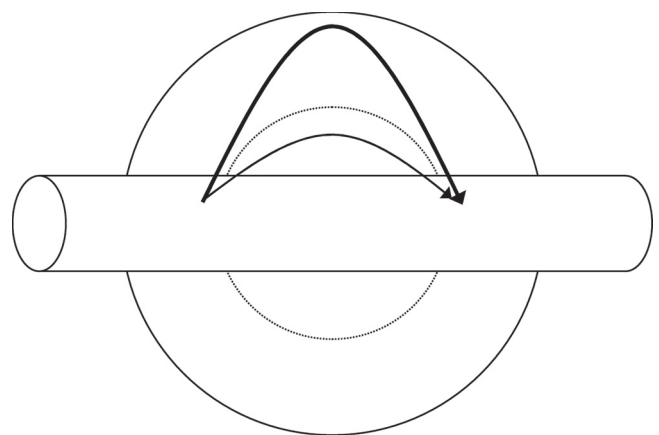


Figure 16 Diffusion paths from the capillary to tissue for rapid, complete diffusive equilibration (bold arrows, large compartment) and slower, incomplete diffusive equilibrium (faint arrows, dashed compartment).

is scaled relative to the volume of blood flow. For an inert gas, therefore, Eq. (18) becomes

$$\frac{\dot{Q}}{V_{ti}\lambda} = \frac{1}{\bar{T}} = \frac{m_0}{\int_0^\infty m(t)dt} \quad (19)$$

Measurement using hyperpolarized noble gases and magnetic resonance imaging

In magnetic resonance imaging (MRI), the source of the signal is the proton spin of native hydrogen nuclei. Conventional ^1H MRI is therefore unable to image the contents of the airway and lung airspaces because they have such a low nuclear spin density, as they contain mainly nitrogen, oxygen, and carbon dioxide. In addition, the numerous air-water interfaces within the lung create magnetic field inhomogeneities that degrade the image. However, if noble gases such as He or Xe are “optically pumped” by circularly polarized laser light, they become hyperpolarized (HP), that is, magnetized by a factor of 10^5 above their native value, and so become an effective contrast agent, visible on MRI. Albert and Cates were the first to demonstrate MRI imaging of the mouse lung by using HP ^{129}Xe (1), and the first images of human airspaces by using ^3He appeared in 1996 (51). While ^3He is highly diffusible, its low blood solubility means that its absorption and conductance to the tissue are minimal. ^{129}Xe , however, is moderately soluble in blood and so is potentially a useful tool to image peripheral tissue perfusion, using washout or distribution models similar to those employed for radioactive tracers as discussed. Despite this favorable property of Xe, it presents a number of other difficulties in practice and for analysis.

First, the magnetic moment of ^{129}Xe (which determines the attainable signal-to-noise ratio) is only about one-third that of ^3He . In addition, the maximum attainable degree of polarization of ^{129}Xe is only about 20% (c.f. 50% for ^3He). This means that the signal will be relatively weak. Swanson et al. were amongst the first to image relative cerebral perfusion by using inhaled ^{129}Xe (72) in the rat. To achieve a sufficient signal, however, it was necessary to ventilate the rat with 100% Xe, which not only induced global hypoxia (itself affecting cerebral perfusion) but also somewhat limits this particular adaptation of the technique for human use. To overcome this “dosing” problem, Duhamel and colleagues used an intra-arterial injection of ^{129}Xe dissolved in a lipid emulsion to deliver a larger bolus dose to the rat brain (18).

The second drawback of HP gases is that the magnetization of ^{129}Xe is transient. Unlike the constantly renewable magnetization of native hydrogen nuclei generated by the pulsed radiofrequency (RF) magnetic field in conventional MRI, the pulsed RF has little energizing effect on HP gas whose magnetization must be achieved by optical pumping before it is inhaled or injected. This magnetization of HP gas is therefore nonrenewable, and the energy of the initial polarization dissipates over a short period of time through T_1

relaxation. T_1 relaxation time is the time constant of the process whereby the longitudinal magnetization vector returns to its equilibrium value. T_1 values are not immutable constants like isotope half-lives but are affected by the experimental conditions. They differ between tissues and are reduced both by RF pulsing and by the ambient Po_2 (this latter property may be exploitable to image the oxygenation state within the airways). In the rat brain, T_1 values are around 16 s (86). This is just about long enough for data acquisition after a bolus injection, but the disappearance of the signal through T_1 relaxation (as opposed to via dilution/washout) must be accounted for in the analysis. MRI/HP gas techniques provide blood flow data that are qualitative and spatially relative rather than “absolute.”

Measurement of perfusion using forced inert gas tension oscillations

The concept of a first-order filter

In the Models and Algebraic Analysis section, we considered the change in the tissue partial pressure of an inert gas if the arterial partial pressure were suddenly changed from zero to some constant value. We also considered the change in tissue partial pressure if the arterial partial pressure were suddenly changed from this constant value to zero. We noted that the kinetics of inert gas wash-in and washout were not necessarily symmetrical. We shall now consider the effect of applying a sinusoidally varying arterial inert gas partial pressure “signal” to a tissue (or an array of tissues perfused in parallel).

If we assume that the tissue acts as a first-order filter, we can expect that the inert gas partial pressure in the tissue (and venous effluent) will vary sinusoidally and that its amplitude and phase will differ from the arterial sinewave. For proof of this, see Gavaghan and Hahn (31). Because the gas is inert, none will be consumed and so the mean arterial and tissue/venous values will be equal. An example of this relationship is shown in Figure 17.

Considering a single tissue element as we did in the Models and Algebraic Analysis section, the mass balance equation is exactly as described in Eq. (2) (except that P_a is now time variant), which is rewritten below:

$$\dot{Q} S_b (P_a(t) - P_v(t)) = V_{ti} S_{ti} \frac{dP_v(t)}{dt} \quad (20)$$

Equation (20) rearranges to

$$\frac{1}{\lambda\tau} (P_a(t) - P_v(t)) = \frac{dP_v(t)}{dt} \quad (21)$$

where, as before, λ is the tissue-blood partition coefficient and τ is the time constant of the system and equal to $V_{ti}/\dot{Q}$.

$P_a(t)$ and $P_v(t)$ are now replaced by sinusoidal functions with identical angular velocities (ω) but different amplitudes (ΔP_a and ΔP_v , respectively) and a phase lag (Φ). Hence,

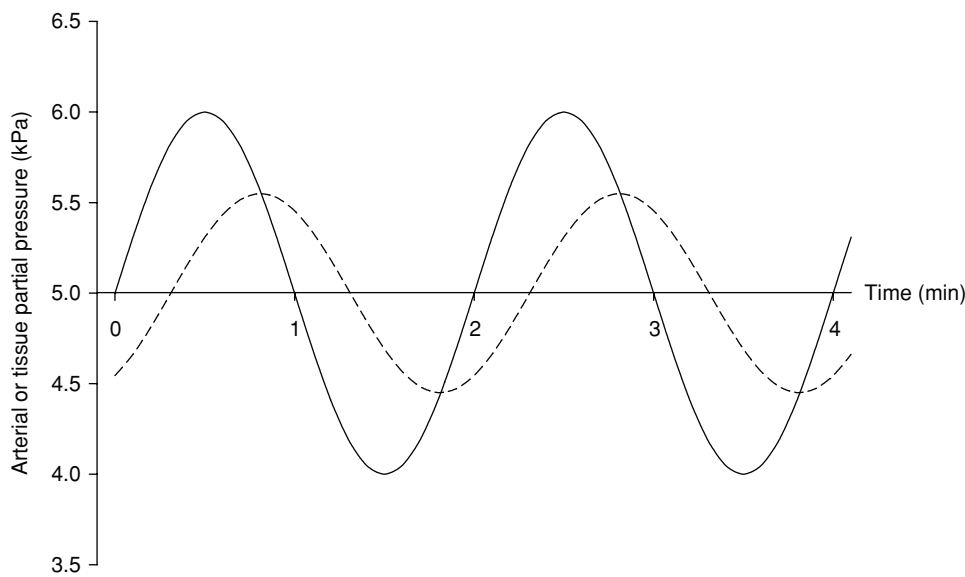


Figure 17 The tissue response (dashed line) to a sinusoidally varying arterial inert gas partial pressure (solid line).

Eq. (21) becomes:

$$\frac{1}{\lambda\tau} [(\bar{P}_a + \Delta P_a \sin(\omega t + \Phi)) - (\bar{P}_v + \Delta P_v \sin(\omega t))] = \frac{d(\bar{P}_v + \Delta P_v \sin(\omega t))}{dt} \quad (22)$$

In the steady state, the mean values, $\bar{P}_a$ and $\bar{P}_v$, are equal and so cancel. The right-hand side of Eq. (22) is differentiated to yield a cosine term.

$$\frac{1}{\lambda\tau} [(\Delta P_a \sin(\omega t + \Phi)) - (\Delta P_v \sin(\omega t))] = \Delta P_v \omega \cos(\omega t) \quad (23)$$

The term $\Delta P_a \sin(\omega t + \Phi)$ in Eq. (23) can be expanded using a standard trigonometric identity “ $\Delta P_a \sin(\omega t + \Phi) = b \sin(\omega t) + a \cos(\omega t)$,” where

$$\Delta P_a^2 = a^2 + b^2 \quad (24)$$

and

$$\Phi = \tan^{-1} \left(\frac{a}{b} \right) \quad (25)$$

Equation (23) can now be written:

$$\frac{1}{\lambda\tau} [b \sin(\omega t) + a \cos(\omega t) - \Delta P_v \sin(\omega t)] = \Delta P_v \omega \cos(\omega t) \quad (26)$$

Comparing the coefficients of the sine terms, we get:

$$b = \Delta P_v$$

Comparing the coefficients of the cos terms, we get:

$$a = \lambda\omega\tau P_v$$

Substituting the terms a and b in Eq. (24) gives:

$\Delta P_a^2 = (\lambda\omega\tau \Delta P_v)^2 + (\Delta P_v)^2$, which simplifies to:

$$\frac{\Delta P_v}{\Delta P_a} = \sqrt{\frac{1}{1 + \omega^2 \tau^2 \lambda^2}} \quad (27)$$

Equation (27) describes how the amplitude of the tissue (and venous) partial pressure signal is damped relative to the input (arterial) signal. Note that the amplitude diminishes as the frequency, the time constant, and the tissue-blood partition coefficient all increase.

The arterial inert gas concentration can be forced to oscillate by introducing a partial pressure oscillation in the inspired gas. If the alveolar partial pressure is taken to be an approximation of the arterial value, and measurements are made in the venous effluent, the amplitude ratio can be used to calculate the blood flow per unit volume (i.e., $1/\tau$) for a given frequency and partition coefficient from Eq. (27). Like the Kety-Schmidt technique, this example of the sinusoid amplitude technique suffers from the drawback of requiring frequent venous sampling. One advantage of this technique is that since the form of the recovered signal is known (i.e., it is sinusoidal and of known frequency), it is easily extracted from measurement noise by using phase-locking analysis techniques.

Note also from Eq. (25) that the phase delay, Φ , is given by:

$$\Phi = \tan^{-1}(\lambda \omega \tau) \quad (28)$$

So measurement of this phase delay also allows the calculation of the blood flow per unit volume of tissue (assuming knowledge of the partition coefficient and the frequency). This is particularly useful because it obviates the need for calibrated measurements of either alveolar or tissue venous partial pressures, since the phase angle can be determined from uncalibrated measurements of these variables.

Whole-body measurements

One of the earliest pioneers of the sinusoid amplitude technique was Zwart, who initially used it as a pharmacological tool to investigate the kinetic properties of the biophase in which volatile anesthetics were thought to exert their effect (90). He later used it as a technique to measure the mean $\dot{V}/\dot{Q}$ ratio in the lung by considering the alveolar unit as a system (91). Here, the input signals are the inspired gas tension sinusoid and any mixed venous signal and the output signals are the alveolar and pulmonary capillary gas tension sinusoids. Analysis of these signals would yield information on the kinetics of the first-order filter, that is, the blood flow per unit ventilation and the ventilation per unit volume of the cardiopulmonary unit. The basic theory of this particular first-order filter is considered as follows.

Consider a homogenous single lung of volume, V_A , which is ventilated by a continuous gas flow of $\dot{V}_A$ and with perfusion $\dot{Q}_P$. The mass balance equation for this system is a little more complex than that of the single tissue element shown in Eq. (20), as it includes contributions from mixed venous, inspired, alveolar, and pulmonary capillary sources. It is shown below:

$$\begin{aligned} \dot{V}_A (P_I(t) - P_A(t)) - S_b \dot{Q}_P (P_c(t) - P_v(t)) \\ = V_A \frac{dP_A}{dt} \end{aligned} \quad (29)$$

Three assumptions are made. The first is that the pulmonary capillary and alveolar partial pressures are equal. The second is that in the steady state, the mean values of the inspired, alveolar, and mixed venous signals are equal and so cancel. The third assumption is that the oscillatory component of the mixed venous signal is small and can be disregarded. This third assumption is important because it means that the oscillatory alveolar signal has no “recirculation” effect. The mixed venous signal therefore does not need to be measured. Gavaghan and Hahn justified this assertion by modeling the size of the mixed venous sinusoidal signal and its effect on the model solution and showed this to be negligible (31). It can be seen from Eq. (27) that if the frequency and partition coefficient are sufficiently high, the mixed venous signal will be attenuated. Venous blood arising from different body com-

partments will also contribute signals with variable phases (due to their different transit times) to the mixed venous blood, resulting in further attenuation of the mixed venous oscillation through destructive interference.

Following a similar sequence of derivation steps as shown in Eqs. (20)–(27), the following expression for the sinusoid amplitude ratio between inspired and expired gas tension can be derived from the mass balance equation:

$$\frac{\Delta P_A}{\Delta P_I} = \sqrt{\frac{1}{\left(1 + S_b \frac{\dot{Q}_P}{\dot{V}_A}\right) + \omega^2 \tau_L^2}} \quad (30)$$

where τ_L is the ventilatory time constant of the lung ($V_A/\dot{V}_A$).

In Eq. (30), there are two unknowns: the mean $\dot{V}/\dot{Q}$ ratio and the ventilatory time constant. Zwart (91) used three arithmetic manipulations to eliminate the unknown, τ_L , to derive the mean ventilation perfusion ratio from the equation.

- i. The first was to consider the alveolar inspired phase lag, Φ . Just as the phase lag was shown to equal $\tan^{-1}(\lambda \tau_L \omega)$ in the single tissue element, Eq. (25), so the phase lag between inspired and alveolar sinusoids in the alveolar system is given by

$$\Phi = \tan^{-1} \left(\frac{\omega \tau_L}{1 + \frac{S_b \dot{Q}_P}{\dot{V}_A}} \right)$$

This rearranges to

$$\omega^2 \tau_L^2 = \tan^2 \Phi \left(1 + \frac{S_b \dot{Q}_P}{\dot{V}_A} \right)^2$$

which when substituted into Eq. (30) gives

$$\frac{\Delta P_A}{\Delta P_I} = \sqrt{\frac{1}{\left(1 + S_b \frac{\dot{Q}_P}{\dot{V}_A}\right) + \tan^2 \Phi \left(1 + S_b \frac{\dot{Q}_P}{\dot{V}_A}\right)^2}} \quad (31)$$

So by measuring the phase lag alone, the $\dot{V}/\dot{Q}$ ratio can be determined, and if ventilation is also known, pulmonary blood flow can be estimated.

- ii. The second was to obtain values for the amplitude ratio ($\Delta P_A/\Delta P_I$) at two different frequencies. The resulting pair of simultaneous equations, in the form of Eq. (30), could then be solved to yield $\dot{V}/\dot{Q}$.
- iii. The third technique was to consider the unique solutions to Eq. (29) at the maxima or minima of the alveolar sinusoid, that is, when $dP_A/dt = 0$. In these conditions, the term V_A (the unknown component of τ_L) is eliminated.

Zwart (91) was able to use this technique, using sub-anesthetic concentrations of halothane, to estimate pulmonary blood flow in healthy dogs. These proved to be reasonably accurate and reproducible when validated by aortic Doppler and the indirect Fick methods.

Gavaghan and Hahn (31) noted that if halothane ($S_b = 2.4$) is used, the left-hand side of the denominator of Eq. (30) dominates even at moderately high frequencies. However, if a less soluble gas such as N_2O ($S_b = 0.47$) is used, then there is greater equipoise between the left- and right-hand sides of the denominator of Eq. (30) and their relative dominance is now governed by ω . At high frequencies, the right-hand side dominates and the effects of pulmonary flow and solubility are eliminated, yielding information on τ_L (and hence V_A , if $\dot{V}_A$ is known). At lower frequencies, the left-hand side dominates, yielding information about $\dot{V}/\dot{Q}$ or $\dot{Q}_P$ if $\dot{V}_A$ is known. So careful selection of the sinusoid frequency should allow either volume or perfusion measurement. N_2O at low concentration (4 ± 2 kPa) may be used with periods of between 1 and 4 min.

Diving and Aerospace

Inert gases are of some importance in diving where nitrogen narcosis (or “rapture”) may be a problem in dives at greater than 40 to 50 msw (meters of sea water), equivalent to 5 to 6 bar, and gas bubble formation, particularly with nitrogen, may occur during decompression even after relatively shallow dives of 20 to 30 msw (3–4 bar). Nitrogen accumulates in repeated dives, as the elimination from the body is slower than the uptake. Venous PN_2 levels in Korean ama breath-hold divers have been shown to rise from a baseline of 584 to 635 torr ($t_{1/2} = 6$ min) and to fall back to base level over 3 to 4 h ($t_{1/2} = 36$ min) (64). Helium is often used with oxygen in breathing mixtures to decrease the nitrogen narcosis risk and to reduce the overall density of the gas mixture (and hence the airways resistance) during deeper dives, but such mixtures require even longer decompression times. There have been several good recent reviews of the physiological issues (28,50,56,74).

Decompression is a potential problem for astronauts who “spacewalk,” as the current space suits are pressurized to 30% of the pressure inside the spacecraft. Earthbound studies in decompression chambers have shown that potentially 17% of “spacewalkers” could develop decompression sickness, but there have never been any reports from astronauts (48). In exposed individuals, there is also the potential for a massive depressurization if the space suit develops a leak; theoretically, above approximately 19-km altitude, there will be gas bubble formation with water boiling off in tissues when the ambient pressure is 6.3 kPa (47 mmHg)—the Armstrong limit (5).

Conclusion

Inert gas transport may be well modeled by a relatively simple ODE mathematical model that may be expanded in com-

plexity to reduce the number of assumptions required. This relationship is dependent on the blood flow, volume of the tissue, and the tissue-blood partition coefficient, and the relationship between uptake and elimination for the organ or even the whole body may not be symmetrical. Computer models may also be used to achieve a close approximation to the physiology, and such models are beginning to be used to provide real-time measurements of various physiological variables, using a few noninvasive measurements and real-time computer analysis, to deduce a range of other derived variables compatible with these few measurements. There is much potential to improve the integration of tissue transport models with models of respiration, particularly with respect to tidal breathing and non-steady-state situations, so that continuous noninvasive physiological measurements may be approximated, enabling exercise, hyperbaric, and anesthetic and intensive care situations to be better researched.

One of the main areas in which inert gas transport is important is in the uptake and eventual elimination of anesthetic agents, with their differing responses heavily dependent on their individual blood and tissue solubilities. These relationships are also of importance where gas mixtures have been used as splints (e.g., in ophthalmology), or there are gas-containing cavities that interact with anesthetic agents, and when blockage occurs in the airways eventually leading to atelectasis with the time scale dependent upon the composition of the cavity gas mixture and the gases breathed.

Measurement of inert gas uptake enables organ perfusion to be estimated using venous equilibrium partial pressures and the tissue-blood solubility coefficient as in Eq. (10) or by bolus injection of gas as in Eq. (18). There have also been measurements made using forced oscillations of inert gases as in Eqs. (30) and (31), though such a technique is still somewhat limited by the technology available to force the gas oscillation and to measure the resultant blood or gas changes. There are still developments to be made with respect to the best agents to be used with this technique dependent on the physical characteristics of the available inert gases and the best forcing and measurement technologies. With better integration of more complex lung, and tissue models in the future, real-time physiological measurements may be possible for airway dead space, lung volumes, dynamic ventilation-perfusion relationships, and even some metabolic and disease relationships, without the need to approximate to an equilibrium state.

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